

# Fabius–Rvachev New Frontiers

**Log-concavity, native orthogonal polynomials, Christoffel reconstruction, rational products for  $\pi$ , Gauss–Padé structure, and Legendre–Gaunt determinants**

Research report for Vladimir Reshetnikov’s ProveIt project

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## Abstract

Let  $F$  denote the bounded Fabius function and let  $w = \text{up}$  be Rvachev’s even  $C^\infty$  probability density supported on  $[-1, 1]$ . The current ProveIt corpus already develops exact dyadic arithmetic, the Thue–Morse and infinite-sinc descriptions, moment recurrences, Bell–Bernoulli formulae, Lambert- $W$  endpoint asymptotics, polynomial synthesis by translates of  $\text{up}$ , and Fourier–Legendre expansions. This report audits that corpus and then develops a different, apparently absent layer: the intrinsic convex and spectral geometry of the probability measure

$$d\mu(x) = \text{up}(x) dx.$$

First,  $\mu$  is an infinite convolution of centered uniform laws. Closure of log-concavity under convolution and weak limits therefore proves that  $\text{up}$ , and hence  $F$  on  $[0, 1]$ , is log-concave. Combining this with  $F'(x) = 2F(2x)$  gives the new Turán inequality

$$F(2x)^2 \geq 2F(x)F(4x), \quad 0 < x \leq \frac{1}{4}.$$

For  $g_n = -\log F(2^{-n})$  this becomes the sharp discrete-curvature bound  $g_n - 2g_{n-1} + g_{n-2} \geq \log 2$ .

Second, let  $\pi_n$  be the monic orthogonal polynomials for  $\mu$ ,

$$\pi_{n+1}(x) = x\pi_n(x) - \beta_n\pi_{n-1}(x).$$

The known logarithmic-square endpoint law places  $\mu$  in the Szegő class. Classical OPRL theory then yields

$$\beta_n \longrightarrow \frac{1}{4}, \quad \frac{1}{n} \sum_{\pi_n(\xi)=0} \delta_\xi \implies \frac{dx}{\pi\sqrt{1-x^2}},$$

together with strong norm, ratio, and determinant asymptotics. The associated Christoffel function reconstructs the bump pointwise:

$$\text{up}(x) = \lim_{N \rightarrow \infty} \frac{N\lambda_N(x)}{\pi\sqrt{1-x^2}}, \quad -1 < x < 1.$$

Because every moment is rational,  $\lambda_N(0)$  is rational; since  $\text{up}(0) = 1$ , this gives the canonical rational limit

$$N\lambda_N(0) \longrightarrow \pi.$$

The strong Szegő asymptotic at the symmetry point gives a second, more surprising formula. Every  $\beta_n$  is rational, yet

$$\pi = 2 \prod_{j=1}^{\infty} \frac{\beta_{2j}}{\beta_{2j-1}}.$$

The finite products are rational. They are linked exactly to the Christoffel approximants by a harmonic-mean identity. The same Jacobi data generate the Gauss rules and the diagonal Padé

approximants of the Cauchy transform. Finally, the repository’s exact Legendre coefficients give a finite Legendre–Gaunt Gram matrix whose determinants recover the  $\beta_n$ ; this closes a new exact circuit from dyadic Fabius values through Wigner  $3j$  coefficients to  $\pi$  and back to up.

A fully commented Python program in the archive computes exact initial data and 4,800-digit data through  $\beta_{200}$ . It finds strict increase toward  $1/4$  throughout the tested range and strongly suggests

$$\frac{1}{4} - \beta_n \sim \frac{2}{\pi} \frac{(\log_2 n)^2}{n^2}, \quad \pi - 2 \prod_{j=1}^M \frac{\beta_{2j}}{\beta_{2j-1}} \sim \frac{(\log_2 M)^2}{M^2}.$$

These rate statements and monotonicity are recorded as conjectures, not as proved consequences of the computations.

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### Evidence and novelty boundary

The report is pinned to repository snapshot b863b90ee74ec0405af2d67ad04c61824c3aea00. The recursive TeX tree under Analysis/FabiusFunction/docs was inspected together with its current coverage, audit, archive, and frontier manifests. Canonical consolidated sources and the newest Legendre/Lagrange reports were close-read; generated tables and superseded siblings were cross-checked through their manifests and targeted formula searches. Repository searches for the defining terms and formulas of the new layer—*log-concavity of up*, native Szegő/Nevai asymptotics, Christoffel reconstruction, the alternating Jacobi product for  $\pi$ , and the Legendre–Gaunt determinant bridge—returned no matching result at the audited snapshot. “New” therefore means new relative to this repository snapshot, not a claim of worldwide priority. Every imported fact, new proof, standard external theorem, numerical observation, and conjecture is marked according to its status.

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### Most important new statements

1. up and  $F|_{[0,1]}$  are log-concave; hence a dyadic Turán inequality, strong discrete convexity of  $-\log F(2^{-n})$ , and convexity of the inverse in logarithmic probability coordinates.
  2. The native up-law Jacobi coefficients satisfy  $\beta_n \rightarrow 1/4$ ; zeros converge to the arcsine law; the moment Hankel determinants have a sharp capacity-scale asymptotic with an explicit Fabius–Szegő constant.
  3. Christoffel functions reconstruct up from exact rational moment matrices, and  $N\lambda_N(0) \rightarrow \pi$  through rational numbers.
  4.  $\pi = 2 \prod_{j \geq 1} \beta_{2j}/\beta_{2j-1}$ , with an exact harmonic-mean relation between its partial products and the Christoffel approximants.
  5. The Cauchy transform has the Jacobi  $J$ -fraction whose convergents are exact Gauss–Padé approximants, while the Fourier–Legendre coefficients produce the same Jacobi data through finite rational Gaunt matrices.
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## Contents

|          |                                                 |          |
|----------|-------------------------------------------------|----------|
| <b>1</b> | <b>Corpus audit and choice of frontier</b>      | <b>5</b> |
| 1.1      | Snapshot and source strata . . . . .            | 5        |
| 1.2      | What the corpus already contains . . . . .      | 5        |
| 1.3      | Why this direction avoids duplication . . . . . | 6        |

|           |                                                                        |           |
|-----------|------------------------------------------------------------------------|-----------|
| <b>2</b>  | <b>Repository baseline and notation</b>                                | <b>6</b>  |
| 2.1       | The Fabius and up laws . . . . .                                       | 6         |
| 2.2       | MGF, cumulants, and rational moments . . . . .                         | 7         |
| 2.3       | Endpoint law . . . . .                                                 | 8         |
| <b>3</b>  | <b>A new log-concavity theorem</b>                                     | <b>8</b>  |
| 3.1       | Infinite uniform convolution . . . . .                                 | 8         |
| 3.2       | Dyadic ratio monotonicity and a Turán inequality . . . . .             | 8         |
| 3.3       | Strong discrete curvature at dyadic points . . . . .                   | 9         |
| 3.4       | The inverse in logarithmic probability coordinates . . . . .           | 10        |
| <b>4</b>  | <b>The native orthogonal-polynomial system</b>                         | <b>10</b> |
| 4.1       | Hankel determinants, norms, and Jacobi coefficients . . . . .          | 10        |
| 4.2       | Low-degree certificates . . . . .                                      | 12        |
| 4.3       | Three equivalent exact engines . . . . .                               | 12        |
| <b>5</b>  | <b>Szegő–Nevai asymptotics forced by endpoint flatness</b>             | <b>13</b> |
| 5.1       | Verification of the Szegő condition . . . . .                          | 13        |
| 5.2       | Nevai limit, ratio asymptotics, and zeros . . . . .                    | 14        |
| 5.3       | Strong norm asymptotics and a Fabius–Szegő constant . . . . .          | 14        |
| <b>6</b>  | <b>Christoffel reconstruction from rational moments</b>                | <b>16</b> |
| 6.1       | Moment-matrix formula and exact arithmetic . . . . .                   | 16        |
| 6.2       | Bulk reconstruction theorem . . . . .                                  | 16        |
| 6.3       | A moment reconstruction of the Fabius function itself . . . . .        | 17        |
| <b>7</b>  | <b>An alternating rational Jacobi product for <math>\pi</math></b>     | <b>17</b> |
| 7.1       | Strong center asymptotics . . . . .                                    | 18        |
| 7.2       | Hankel-determinant form . . . . .                                      | 18        |
| 7.3       | Exact harmonic mean of the product approximants . . . . .              | 19        |
| 7.4       | Initial rational products . . . . .                                    | 19        |
| <b>8</b>  | <b>Gauss quadrature and Padé approximation of the Cauchy transform</b> | <b>19</b> |
| 8.1       | Exact Gaussian rules . . . . .                                         | 20        |
| 8.2       | The Jacobi continued fraction . . . . .                                | 20        |
| 8.3       | Coupling to dyadic renormalization . . . . .                           | 22        |
| <b>9</b>  | <b>The Legendre–Gaunt determinant bridge</b>                           | <b>22</b> |
| 9.1       | Exact Legendre data . . . . .                                          | 22        |
| 9.2       | Finite Gaunt expansion . . . . .                                       | 22        |
| 9.3       | Determinants recover the native recurrence . . . . .                   | 23        |
| 9.4       | Christoffel functions directly in the Legendre basis . . . . .         | 24        |
| <b>10</b> | <b>Numerical experiment and exact verification</b>                     | <b>24</b> |
| 10.1      | Two arithmetic modes . . . . .                                         | 24        |
| 10.2      | Observed recurrence coefficients . . . . .                             | 25        |
| 10.3      | Observed product errors . . . . .                                      | 25        |
| 10.4      | Reproducibility and data products . . . . .                            | 26        |

|                                                                      |           |
|----------------------------------------------------------------------|-----------|
| <b>11 Conjectures suggested by the spectral data</b>                 | <b>27</b> |
| 11.1 Monotonicity and one-sided rational approximants . . . . .      | 27        |
| 11.2 A logarithmic-square correction at the Nevai edge . . . . .     | 27        |
| 11.3 Dyadic periodic modulation . . . . .                            | 28        |
| 11.4 Strict log-concavity and strict Turán inequalities . . . . .    | 28        |
| 11.5 The central Christoffel shift . . . . .                         | 28        |
| <b>12 A geometric <math>q</math>-uniform extension</b>               | <b>29</b> |
| 12.1 Definition, transform, and cumulants . . . . .                  | 29        |
| 12.2 Log-concavity and the Szegő class . . . . .                     | 29        |
| 12.3 A continuum of Jacobi products . . . . .                        | 30        |
| <b>13 Research program and formalization roadmap</b>                 | <b>30</b> |
| 13.1 Prove monotonicity by a convolution flow . . . . .              | 30        |
| 13.2 Derive the log-square Jacobi correction . . . . .               | 30        |
| 13.3 Christoffel–Lambert double scaling . . . . .                    | 31        |
| 13.4 Exploit the Padé renormalization residual . . . . .             | 31        |
| 13.5 Arithmetic of Hankel and Gaunt determinants . . . . .           | 31        |
| 13.6 The geometric $q$ -surface . . . . .                            | 32        |
| 13.7 Numerical improvements . . . . .                                | 32        |
| 13.8 Lean formalization order . . . . .                              | 32        |
| <b>14 Conclusion</b>                                                 | <b>33</b> |
| <b>A A detailed small-box estimate for the <math>q</math>-family</b> | <b>33</b> |
| <b>B Exact low-degree spectral table</b>                             | <b>34</b> |
| <b>C Normalization checks against classical weights</b>              | <b>35</b> |
| C.1 Lebesgue weight . . . . .                                        | 35        |
| C.2 Uniform probability weight and the center constant . . . . .     | 35        |
| <b>D Computational method in pseudocode</b>                          | <b>35</b> |
| <b>E Corpus-audit notes</b>                                          | <b>36</b> |

# 1 Corpus audit and choice of frontier

## 1.1 Snapshot and source strata

The branch moved while this investigation was in progress, so all statements are deliberately pinned to b863b90ee74ec0405af2d67ad04c61824c3aea00, whose commit time is 30 August 2026. The source tree is not a single paper. It has several strata:

1. the current proof-backed exposition `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`;
2. the current mathematical glossary and the non-elementarity report;
3. vendored source papers, especially Arias de Reyna’s 2017 analytic and arithmetic papers;
4. a consolidated semi-formalized frontier volume and thematic drafts on asymptotics, inverse functions,  $q$ -calculus, interpolation, integral transforms, spectral arithmetic, and representation theory;
5. recent Legendre and Lagrange closed-loop reports, including `legendre_rvachev_closed_loop.tex` and `legendre_rvachev_self_reconstruction.tex`;
6. an archive of superseded expositions and early calculations; and
7. generated TeX tables that are inputs to larger reports rather than independent mathematical narratives.

The coverage and audit documents are essential because the archive contains near-duplicates with different correction levels. Treating every archived copy as independent would overcount evidence and, more seriously, could revive claims explicitly corrected in the canonical exposition.

## 1.2 What the corpus already contains

The following list is a compressed dependency map, not a substitute for the repository’s much longer crosswalk.

- The bounded CDF  $F$ , the signed/global variants, the even bump up, the refinement and differential equations, support, smoothness, endpoint flatness, and partition-of-unity identities.
- The weighted-uniform random series, finite convolutions, Fourier and MGF products, the Thue–Morse product, Poisson summation, and exact dyadic values.
- Rational moments and their triangular recurrence, cumulants expressed by Bernoulli numbers, moments expressed by complete Bell polynomials, and positive moment Hankel matrices.
- Sharp Fourier decay and log-periodic dyadic-shell structure, together with the small-argument endpoint law and its Lambert- $W$  inversion.
- $q$ -Pochhammer and Gaussian-binomial identities, geometric Richardson filters, and several exact finite interpolation/sampling mechanisms.
- Fourier–Legendre expansions of up, finite synthesis of polynomials by shifted copies of up, and the resulting blockwise Legendre–Rvachev self-reconstruction.
- Native orthogonal-polynomial data: exact moments and positive Hankel matrices, together with an all-degree exact rational layer for Hankel determinants, monic polynomial coefficients, norms, zero diagonal, positive Jacobi subdiagonals, orthogonality, determinant cross-ratio, and the centered recurrence; the separately evaluated low-degree table has the formal/report boundary stated in Section 4.

- The moment Laurent series and a renormalization identity for the Cauchy transform.

The last two bullets are where this report begins. The repository had built the finite exact algebraic layer, but its own canonical discussion still identified the full Jacobi/Padé theory as undeveloped. No audited TeX source used Szegő, Nevai, Rakhmanov, Christoffel, Gauss quadrature, or the central Jacobi product developed below.

### 1.3 Why this direction avoids duplication

The recent Legendre reports solve a different closure problem. They begin with an exact expansion of up in the *classical* Legendre basis and replace each Legendre polynomial by a finite train of translated up-functions. Here we instead orthogonalize polynomials against the *native measure*  $\text{up}(x) dx$ . The resulting Jacobi matrix is an invariant of the probability law itself. It remembers all rational moments, controls Gaussian quadrature and Padé approximation, and has a universal large-degree limit fixed by the support. The two viewpoints meet only later, through the finite Gaunt matrix of [section 9](#).

Table 1: Claim-status summary. “Repository baseline” means imported from b863b90ee74ec0405af2d67ad04c61824c3aea00; “standard theorem” means a classical external result whose hypotheses are verified here.

| Statement                                                     | Status                   | Main input                                                 |
|---------------------------------------------------------------|--------------------------|------------------------------------------------------------|
| Uniform-series law, rational moments, endpoint log-square law | Repository baseline      | Canonical exposition and formal moment modules             |
| up and $F$ are log-concave                                    | New proof here           | Convolution/weak-limit closure of log-concavity            |
| Dyadic Turán inequality and action curvature                  | New theorem here         | Log-concavity plus $F'(x) = 2F(2x)$                        |
| $\beta_n \rightarrow 1/4$ , arcsine zeros                     | New application here     | Rakhmanov/regular OPRL theory                              |
| Fabius–Szegő norm constant                                    | New application here     | Strong Szegő norm asymptotics                              |
| Christoffel reconstruction of up                              | New application here     | Máté–Nevai–Totik/Totik asymptotics                         |
| Rational limits to $\pi$                                      | New corollary here       | $\text{up}(0) = 1$ and rational moments                    |
| Alternating Jacobi product for $\pi$                          | New theorem here         | Strong Szegő center asymptotic                             |
| Gauss–Padé package                                            | New synthesis here       | Classical Markov/Stieltjes theory                          |
| Legendre–Gaunt determinant bridge                             | New finite identity here | Existing exact Legendre coefficients                       |
| Monotone $\beta_n$ , precise log-square rate                  | Conjectural              | Exact checks and 4,800-digit computation through $n = 200$ |

## 2 Repository baseline and notation

### 2.1 The Fabius and up laws

Let  $U_0, U_1, \dots$  be independent uniform random variables on  $[0, 1]$ , and put

$$X = \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}}. \quad (2.1)$$

The bounded Fabius function is the distribution function

$$F(x) = \Pr[X \leq x]. \quad (2.2)$$

It is supported probabilistically on  $[0, 1]$ , is strictly increasing there, and obeys

$$F'(x) = 2F(2x), \quad 0 \leq x \leq \frac{1}{2}, \quad (2.3)$$

with the usual reflected form on the other half. Rvachev's bump is

$$\text{up}(y) = F(1 - |y|), \quad |y| \leq 1, \quad (2.4)$$

and vanishes outside  $[-1, 1]$ . It is even,  $C^\infty$ , positive on  $(-1, 1)$ , flat at the endpoints, and normalized by

$$\int_{-1}^1 \text{up}(x) \, dx = 1, \quad \text{up}(0) = 1. \quad (2.5)$$

The centered random variable

$$Y = 2X - 1 = \sum_{n=0}^{\infty} 2^{-n} (U_n - \frac{1}{2}) \quad (2.6)$$

has density  $\text{up}$ . Indeed, the density of  $X$  is  $F'(x)$ ; for  $-1 < y < 0$ ,

$$\frac{1}{2} F' \left( \frac{y+1}{2} \right) = F(y+1) = \text{up}(y),$$

and symmetry gives the positive half. Thus throughout the report

$$d\mu(x) = \text{up}(x) \, dx \quad (2.7)$$

is both a compactly supported probability measure and an infinite convolution of centered uniforms.

## 2.2 MGF, cumulants, and rational moments

The moment-generating function of  $Y$  is the entire product

$$M(z) = \int_{-1}^1 e^{zx} \, d\mu(x) = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j}. \quad (2.8)$$

Writing

$$\log M(z) = \sum_{r \geq 1} \kappa_r \frac{z^r}{r!},$$

all odd cumulants vanish and

$$\kappa_{2r} = \frac{2^{2r-1} B_{2r}}{r(2^{2r} - 1)}. \quad (2.9)$$

Consequently every moment is rational and can be written as a complete Bell polynomial in the Bernoulli cumulants. For the even moments

$$c_n = m_{2n} = \int_{-1}^1 x^{2n} \text{up}(x) \, dx, \quad m_{2n+1} = 0, \quad (2.10)$$

the particularly efficient triangular recurrence is

$$(2n+1)(2^{2n} - 1)c_n = \sum_{j=0}^{n-1} \binom{2n+1}{2j} c_j, \quad c_0 = 1. \quad (2.11)$$

The first values are

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}. \quad (2.12)$$

### 2.3 Endpoint law

The sharp asymptotic program in the repository implies, in its simplest invariant form,

$$-\log F(x) \sim \frac{\log^2(1/x)}{2 \log 2} \quad (x \downarrow 0). \quad (2.13)$$

Therefore

$$-\log \text{up}(1 - \varepsilon) \sim \frac{\log^2(1/\varepsilon)}{2 \log 2} \quad (\varepsilon \downarrow 0), \quad (2.14)$$

with the symmetric law at  $-1$ . More refined repository formulae introduce a Lambert- $W$  saddle coordinate and dyadic log-periodic corrections. For the spectral results below, the leading logarithmic-square envelope is enough.

## 3 A new log-concavity theorem

### 3.1 Infinite uniform convolution

For  $n \geq 0$ , let  $\nu_n$  be the uniform probability law on

$$I_n = [-2^{-n-1}, 2^{-n-1}].$$

Then [equation \(2.6\)](#) says

$$\mu = \nu_0 * \nu_1 * \nu_2 * \cdots \quad (3.1)$$

in the sense of weak convergence of the finite convolutions.

**Theorem 3.1** (Log-concavity of the up law). *The probability measure  $\mu$  is log-concave. Its continuous density up is log-concave on  $\mathbb{R}$  in the extended sense, and strictly positive on  $(-1, 1)$ . Consequently  $F$  is log-concave on  $[0, 1]$ .*

*Proof.* The indicator of an interval is log-concave in the extended sense, so every  $\nu_n$  is a log-concave probability measure. The Prékopa–Leindler theorem [3] implies that convolution preserves log-concavity. Hence every finite convolution  $\mu_N = \nu_0 * \cdots * \nu_N$  is log-concave. The class of log-concave probability measures is closed under weak convergence, so the weak limit  $\mu$  is log-concave. Since  $\mu$  has the continuous full-dimensional density up on its support, that density is log-concave. Finally, [equation \(2.4\)](#) gives  $F(x) = \text{up}(x - 1)$  for  $0 \leq x \leq 1$ , and translations preserve log-concavity.  $\square$

*Remark 3.2.* This proof uses only the probability representation; it is independent of the Fourier product, exact dyadic arithmetic, or endpoint asymptotics. It also extends verbatim to the geometric-uniform family in [section 12](#).

### 3.2 Dyadic ratio monotonicity and a Turán inequality

Because  $F > 0$  on  $(0, 1]$ , log-concavity is equivalent to concavity of  $\log F$ . Combining this with the differential equation creates a nonlinear inequality that does not appear in the defining linear refinement law.

**Theorem 3.3** (Dyadic ratio monotonicity). *The function*

$$R(x) = \frac{F(2x)}{F(x)} \quad (3.2)$$

*is nonincreasing on  $(0, \frac{1}{2}]$ .*



*Proof.* By [equation \(2.3\)](#),

$$(\log F)'(x) = \frac{F'(x)}{F(x)} = 2 \frac{F(2x)}{F(x)} = 2R(x).$$

The derivative of a differentiable concave function is nonincreasing.  $\square$

**Corollary 3.4** (Fabius Turán inequality). *For  $0 < x \leq \frac{1}{4}$ ,*

$$\boxed{F(2x)^2 \geq 2F(x)F(4x)}. \quad (3.3)$$

*Every dyadic specialization of [equation \(3.3\)](#) is an exact inequality between rational numbers.*

*Proof.* Differentiate [equation \(3.2\)](#) for  $0 < x < 1/4$  and use  $F'(x) = 2F(2x)$  twice:

$$R'(x) = \frac{4F(4x)F(x) - 2F(2x)^2}{F(x)^2}.$$

By [theorem 3.3](#),  $R'(x) \leq 0$ , which is exactly [equation \(3.3\)](#). Continuity includes the endpoint  $x = 1/4$ .  $\square$

### 3.3 Strong discrete curvature at dyadic points

Define

$$a_n = F(2^{-n}), \quad g_n = -\log a_n \quad (n \geq 0). \quad (3.4)$$

Thus  $g_0 = 0$  and  $g_1 = \log 2$ .

**Corollary 3.5** (Sharp discrete-curvature lower bound). *For every  $n \geq 2$ ,*

$$g_n - 2g_{n-1} + g_{n-2} \geq \log 2. \quad (3.5)$$

*Equivalently,*

$$r_n := g_n - \frac{\log 2}{2}n(n+1) \quad (3.6)$$

*is a convex sequence. In particular,*

$$F(2^{-n}) \leq 2^{-n(n+1)/2}. \quad (3.7)$$

*The curvature constant  $\log 2$  is asymptotically sharp because of [equation \(2.13\)](#).*

*Proof.* Set  $x = 2^{-n}$  in [equation \(3.3\)](#):

$$a_{n-1}^2 \geq 2a_n a_{n-2}.$$

Taking negative logarithms gives [equation \(3.5\)](#). The quadratic in [equation \(3.6\)](#) has second difference  $\log 2$ , so the residual is convex. Comparing with the unique sequence having the same initial values and constant second difference  $\log 2$  proves [equation \(3.7\)](#). Finally, [equation \(2.13\)](#) gives  $g_n \sim (\log 2)n^2/2$ , whose second difference tends to  $\log 2$  at leading order.  $\square$

### 3.4 The inverse in logarithmic probability coordinates

Let  $F^{-1} : (0, 1) \rightarrow (0, 1)$  denote the inverse of the strictly increasing CDF. The Lambert- $W$  analysis in the repository studies this inverse near zero. Log-concavity supplies a global shape constraint in the natural exponential coordinate.

**Theorem 3.6** (Convex logarithmic quantile). *The function*

$$q(s) = F^{-1}(e^{-s}), \quad s > 0, \quad (3.8)$$

*is decreasing and convex.*

*Proof.* Set  $h(x) = \log F(x)$ . By [theorem 3.1](#),  $h$  is increasing and concave. The inverse of an increasing concave function is increasing and convex. Thus  $h^{-1}(t) = F^{-1}(e^t)$  is convex in  $t < 0$ . Composing with the affine map  $t = -s$  preserves convexity and reverses monotonicity.  $\square$

*Remark 3.7.* Theorem 3.6 is a useful admissibility test for any truncated Lambert- $W$  expansion or rational approximation to  $F^{-1}$ . An approximation that develops a negative second derivative in  $s$  outside its explicit error bar cannot be globally faithful.

## 4 The native orthogonal-polynomial system

The classical Legendre polynomials are orthogonal for Lebesgue measure on  $[-1, 1]$ . The probability density  $w = \text{up}$  defines a second, intrinsic orthogonal system. It is this system, rather than the Legendre expansion of  $w$  itself, that carries the moment problem, Gaussian quadrature, and the continued fraction of the Cauchy transform.

### 4.1 Hankel determinants, norms, and Jacobi coefficients

Write

$$m_k = \int_{-1}^1 x^k w(x) dx, \quad \Delta_N = \det[m_{j+k}]_{j,k=0}^{N-1}, \quad \Delta_0 = 1. \quad (4.1)$$

The repository proves  $\Delta_N > 0$  for every  $N$ : a nonzero polynomial has only finitely many roots, while  $w$  is positive throughout  $(-1, 1)$ . Hence there is a unique monic polynomial  $\pi_n$  of degree  $n$  orthogonal to all lower powers. Put

$$H_n = \int_{-1}^1 \pi_n(x)^2 w(x) dx. \quad (4.2)$$

The standard Gram-determinant argument gives

$$H_n = \frac{\Delta_{n+1}}{\Delta_n}. \quad (4.3)$$

Evenness of  $w$  forces  $\pi_n(-x) = (-1)^n \pi_n(x)$  and removes the diagonal Jacobi parameters. With  $\pi_0 = 1$  and  $\pi_1 = x$ ,

$$\boxed{\pi_{n+1}(x) = x\pi_n(x) - \beta_n\pi_{n-1}(x)}, \quad \beta_n = \frac{H_n}{H_{n-1}} = \frac{\Delta_{n+1}\Delta_{n-1}}{\Delta_n^2} > 0. \quad (4.4)$$

**Proposition 4.1** (Exact arithmetic of the native Jacobi matrix). *Every coefficient of every  $\pi_n$ , every squared norm  $H_n$ , every Hankel determinant  $\Delta_n$ , and every recurrence coefficient  $\beta_n$  is rational.*

*Proof.* The recurrence [equation \(2.11\)](#) gives  $m_k \in \mathbb{Q}$ . Thus  $\Delta_n \in \mathbb{Q}$ . Positivity makes every division in [equations \(4.3\) and \(4.4\)](#) legitimate, proving  $H_n, \beta_n \in \mathbb{Q}$ . Starting from 1,  $x$  in [equation \(4.4\)](#) then proves coefficientwise rationality of  $\pi_n$  by induction.  $\square$

*Remark 4.2* (Current Lean boundary and complete declaration inventory). The scalar-base-change module `GramStieltjesNaturality.lean` contains zero public definitions and exactly six public theorems: `momentPairing_map`, `map_gramStieltjesNumerator`, `map_gramStieltjesPolynomial`, `map_gramStieltjesNorm`, `map_gramStieltjesJacobiDiagonal`, and `map_gramStieltjesJacobiSubdiagonal`. These are finite-algebra naturality results: the numerator theorem is ring-level, while normalization and Jacobi quotients use fields; there is no measure, positivity, or convergence hypothesis.

The native specialization `RvachevRationalJacobi.lean` contains exactly four public definitions—`rvachevHankelDetRat`, `rvachevOrthoPolynomialRat`, `rvachevOrthoNormRat`, and `rvachevJacobiSubdiagonalRat`—and thirteen public theorems: `upMoment_eq_rvachevRawMomentRat_cast`, `rvachevHankelDetRat_cast`, `rvachevHankelDetRat_pos`, `rvachevOrthoPolynomialRat_cast`, `rvachevOrthoPolynomialRat_isMonicOfDegree`, `momentPairing_rvachevOrthoPolynomialRat_eq_zero`, `rvachevOrthoNormRat_cast`, `rvachevOrthoNormRat_pos`, `gramStieltjesJacobiDiagonal_rvachevRawMomentRat_eq_zero`, `rvachevJacobiSubdiagonalRat_cast`, `rvachevJacobiSubdiagonalRat_pos`, `rvachevJacobiSubdiagonalRat_eq_det_ratio`, and `rvachevOrthoPolynomialRat_thre_e_term`. Together they give, at every degree over  $\mathbb{Q}$ , positive  $\Delta_n$ , every coefficient of monic  $\pi_n$ , positive  $H_n$ , a zero diagonal, positive subdiagonals, exact rational orthogonality, the Hankel cross-ratio, the centered recurrence, and all real-cast bridges. The definition `rvachevJacobiSubdiagonalRat` is zero-based: its value at index  $n$  is the report coefficient  $\beta_{n+1}$ .

These are noncomputable adjugate-based finite definitions, not a rational evaluator. In particular, they do not name or evaluate the explicit  $H_4$  and  $\beta_4$  fractions displayed below. Those two modules by themselves also prove no roots, Christoffel reconstruction or central products, Gauss quadrature, second-kind polynomials, continued-fraction/Padé identification, or Szegő/Nevai/Rakhmanov asymptotics.

The later finite change-of-basis tranche closes precisely the determinant part of [section 9](#). The generic module `PolynomialMomentGramDeterminant.lean` has exactly two public definitions, `polynomialCoefficientMatrix` and `polynomialMomentGramMatrix`, and seven public theorems: `polynomialCoefficientMatrix_apply`, `polynomialMomentGramMatrix_apply`, `polynomialMomentGramMatrix_eq_transpose_mul_hankel_mul`, `polynomialMomentGramMatrix_det_eq_coefficient_det_sq_mul`, `polynomialCoefficientMatrix_det_eq_prod_coeff`, `polynomialMomentGramMatrix_det_eq_prod_coeff_sq_mul`, and `gramStieltjesJacobiSubdiagonal_eq_polynomialMomentGramMatrix_det_ratio`. For a coherent polynomial family, it expands the moment pairing to  $G = C^T H C$ , observes that the degree bound makes  $C$  upper triangular with the leading coefficients on its diagonal, takes determinants, and then transports the zero-based Hankel cross-ratio through the resulting diagonal product. The generic cross-ratio theorem has no Hankel-nonvanishing hypothesis: Lean’s field division is total, so a zero middle Hankel determinant makes both cross-ratios zero. That singular equality is not a genuine nonsingular Jacobi recurrence.

The specialization `FabiusLegendreHankelDeterminant.lean` has exactly two public definitions, `upLegendreGramMatrix` and `upLegendreGramDet`, and seven public theorems: `upLegendreGramMatrix_apply_eq_integral`, `upLegendreGramDet_eq_prod_leadingCoeff_sq_mul_hankelDet`, `upLegendreGramDet_zero`, `upLegendreGramDet_pos`, `coeff_legendrePolynomial_self_div_succ`, `gramStieltjesJacobiSubdiagonal_upMoment_eq_upLegendreGramDet_ratio`, and `rvachevJacobiSubdiagonalRat_cast_eq_upLegendreGramDet_ratio`. It formalizes [equation \(9.7\)](#) and the zero-based form of [equation \(9.8\)](#): its input index  $n$  denotes the report coefficient  $\beta_{n+1}$ , so the prefactor is  $((n+1)/(2n+1))^2$  and the determinant quotient is  $D_{n+2}D_n/D_{n+1}^2$ . For an arbitrary `BoundedFabius` input, Lean proves the determinant identity, the empty-determinant case  $D_0 = 1$ , the Legendre leading-coefficient quotient, and this real Gram cross-ratio. The entry-as-integral identity, strict positivity, and the bridge identifying the ratio with the real cast of the exact rational Rvachev subdiagonal additionally require

IsFabius. Under that hypothesis positivity excludes the totalized singular case.

This closure does *not* formalize the finite Gaunt integral, Wigner/3j-symbol, or factorial entry formulas [equations \(9.3\) to \(9.5\)](#); consequently it does not establish entry rationality by that route. It also proves no Christoffel reconstruction. Finally, the existing `rvachevTranslateGram` is a different object: it pairs shifted-up atoms under unweighted interval integration, whereas `upLegendreGramMatrix` pairs ordinary polynomials under the up-law moment functional.

This is an unusually rigid situation. The infinite Jacobi matrix

$$J = \begin{pmatrix} 0 & \sqrt{\beta_1} & & & \\ \sqrt{\beta_1} & 0 & \sqrt{\beta_2} & & \\ & \sqrt{\beta_2} & 0 & \sqrt{\beta_3} & \\ & & & \ddots & \ddots & \ddots \end{pmatrix} \quad (4.5)$$

has algebraic entries, while its spectral measure at the first basis vector is the non-elementary  $C^\infty$  law  $w(x) dx$ . Thus the analytic complexity of up is encoded in the asymptotic arrangement of exact rational numbers  $\beta_n$ .

## 4.2 Low-degree certificates

The first moments from [equation \(2.11\)](#) are

$$m_0 = 1, \quad m_2 = \frac{1}{9}, \quad m_4 = \frac{19}{675}, \quad m_6 = \frac{583}{59535}. \quad (4.6)$$

The repository directly formalizes  $\pi_0, \dots, \pi_4$  and  $H_0, \dots, H_3$ ; the displayed  $\beta_1, \beta_2, \beta_3$  then follow as exact quotients of those formal norm values. The final-row  $H_4$  and  $\beta_4$  fractions are instead report-level exact computations, not named Lean evaluations. The combined table is included to fix every normalization used later:

$$\begin{array}{lll} \pi_0(x) = 1, & H_0 = 1, & \\ \pi_1(x) = x, & H_1 = \frac{1}{9}, & \beta_1 = \frac{1}{9}, \\ \pi_2(x) = x^2 - \frac{1}{9}, & H_2 = \frac{32}{2025}, & \beta_2 = \frac{32}{225}, \\ \pi_3(x) = x^3 - \frac{19}{75}x, & H_3 = \frac{19808}{7441875}, & \beta_3 = \frac{619}{3675}, \\ \pi_4(x) = x^4 - \frac{62}{147}x^2 + \frac{619}{33075}, & H_4 = \frac{26727424}{55791736875}, & \beta_4 = \frac{835232}{4640643}. \end{array} \quad (4.7)$$

The two final-row fractions follow from the same exact moment algorithm and are independently verified by the archive’s Python program. The all-degree rational objects described above prove their rationality and positivity but do not compute these explicit values. Numerically,

$$\beta_1 = 0.111111\dots, \quad \beta_2 = 0.142222\dots, \quad \beta_3 = 0.168435\dots, \quad \beta_4 = 0.179981\dots \quad (4.8)$$

## 4.3 Three equivalent exact engines

There are three useful ways to compute the same data.

**Proposition 4.3** (Moment, determinant, and recurrence engines). *For every  $n \geq 0$  the following procedures agree.*

1. Form  $\Delta_n$  from the rational moments and use [equation \(4.4\)](#).
2. Solve the lower-triangular orthogonality equations for the coefficients of the monic  $\pi_n$ .

3. Once  $\pi_n$  is known, use the norm-producing moment

$$H_n = \int x^n \pi_n(x) w(x) dx \quad (4.9)$$

and then  $\beta_n = H_n/H_{n-1}$ .

Method 3 uses only  $\mathcal{O}(n^2)$  coefficient operations through degree  $n$ .

*Proof.* The difference between  $x^n$  and  $\pi_n$  has degree below  $n$ , hence is orthogonal to  $\pi_n$ . Therefore  $\langle \pi_n, x^n \rangle = \langle \pi_n, \pi_n \rangle = H_n$ , proving [equation \(4.9\)](#). The determinant and triangular systems are the two standard constructions of the same unique monic orthogonal polynomial. The recurrence then follows from orthogonality and parity.  $\square$

The rational-native Lean tranche formalizes the determinant construction, unique monic orthogonality, norm quotient, determinant cross-ratio, and centered recurrence that form the algebraic core of these engines. It does not package the three procedures as an executable algorithm, prove their operational equivalence as this proposition states it, or certify the  $\mathcal{O}(n^2)$  operation count.

The third engine is numerically preferable to expanding  $\int \pi_n^2 d\mu$ , but it remains severely ill-conditioned in ordinary floating-point arithmetic. [Section 10](#) describes the high-precision safeguards used here.

## 5 Szegő–Nevai asymptotics forced by endpoint flatness

The endpoint of  $w$  is infinitely flat, but it is not too flat for Szegő theory. This is the key observation that turns the repository’s Lambert- $W$ /log-square endpoint analysis into global spectral information.

### 5.1 Verification of the Szegő condition

For an absolutely continuous measure on  $[-1, 1]$ , the Szegő condition is

$$\int_{-1}^1 \frac{\log w(x)}{\sqrt{1-x^2}} dx > -\infty. \quad (5.1)$$

Only the two endpoints require attention.

**Theorem 5.1** (The up-law is in the Szegő class). *The density  $w = \text{up}$  satisfies [equation \(5.1\)](#).*

*Proof.* The density is continuous and strictly positive on every compact subinterval of  $(-1, 1)$ , so  $\log w$  is bounded there. Near  $x = 1$ , put  $t = 1 - x$ . The relation  $w(x) = F(1 - x)$  for  $0 \leq x < 1$  and the endpoint expansion [equation \(2.13\)](#) imply, for sufficiently small  $t > 0$ ,

$$-\log w(1 - t) \leq C(1 + \log^2(1/t)) \quad (5.2)$$

with a finite constant  $C$ . Moreover  $1 - (1 - t)^2 = t(2 - t) \asymp t$ . Hence the negative part of the endpoint integral is bounded by a constant multiple of

$$\int_0^\varepsilon \frac{1 + \log^2(1/t)}{\sqrt{t}} dt < \infty. \quad (5.3)$$

Evenness handles  $x = -1$ .  $\square$

The same proof reveals a useful threshold: any endpoint law  $-\log w(1 - t) = \mathcal{O}(t^{-\alpha})$  satisfies the Szegő condition when  $\alpha < 1/2$ . The Fabius logarithmic square is much smaller than every such power, despite infinite flatness.

## 5.2 Nevai limit, ratio asymptotics, and zeros

We now invoke the Denisov–Rakhmanov theorem and the regularity theory of orthogonal polynomials on an interval; see [6, Chs. 1 and 12], [7, Chs. 1–3], and [4, Ch. 12]. In our normalization the orthonormal off-diagonal coefficient is  $a_n = \sqrt{\beta_n}$ .

**Theorem 5.2** (Native Nevai limit). *For the up-law Jacobi parameters,*

$$\boxed{\lim_{n \rightarrow \infty} \beta_n = \frac{1}{4}.} \quad (5.4)$$

*Equivalently,  $a_n \rightarrow 1/2$  and every diagonal Jacobi parameter is identically zero.*

*Proof.* The measure is supported on  $[-1, 1]$  and has density  $w(x) > 0$  for Lebesgue almost every  $x \in [-1, 1]$ . Rakhmanov’s theorem therefore gives  $a_n \rightarrow 1/2$  and  $b_n \rightarrow 0$ . Symmetry already gives  $b_n = 0$  exactly. Since  $\beta_n = a_n^2$ , [equation \(5.4\)](#) follows.  $\square$

**Corollary 5.3** (Ratio and root asymptotics). *Let*

$$\Phi(z) = z + \sqrt{z^2 - 1}, \quad (5.5)$$

*where the square root is chosen so that  $\sqrt{z^2 - 1} \sim z$  at infinity. Then, locally uniformly on  $\mathbb{C} \setminus [-1, 1]$ ,*

$$\frac{\pi_{n+1}(z)}{\pi_n(z)} \longrightarrow \frac{\Phi(z)}{2}. \quad (5.6)$$

*If  $x_{n,1} < \dots < x_{n,n}$  are the zeros of  $\pi_n$ , then*

$$\frac{1}{n} \sum_{j=1}^n \delta_{x_{n,j}} \Longrightarrow \frac{dx}{\pi \sqrt{1 - x^2}}. \quad (5.7)$$

*Moreover,*

$$H_n^{1/(2n)} \longrightarrow \frac{1}{2}. \quad (5.8)$$

The limiting law in [equation \(5.7\)](#) is independent of the highly nonclassical local shape of  $w$ . It is the equilibrium measure of the support  $[-1, 1]$ . The fifth figure in the archive compares the 100 zeros obtained from the Jacobi truncation with the corresponding arcsine quantiles.

## 5.3 Strong norm asymptotics and a Fabius–Szegő constant

The logarithmic endpoint condition permits the strong norm form of Szegő’s theorem. Define

$$\mathcal{C}_F = 2\pi \exp \left[ \frac{1}{\pi} \int_0^\pi \log(w(\cos \theta) \sin \theta) d\theta \right]. \quad (5.9)$$

The integral is finite by [theorem 5.1](#), and  $\mathcal{C}_F > 0$ .

**Theorem 5.4** (Strong monic norm asymptotic). *The native monic norms satisfy*

$$\boxed{4^n H_n \longrightarrow \mathcal{C}_F.} \quad (5.10)$$

*Consequently,*

$$\mathcal{C}_F = \prod_{n=1}^{\infty} 4\beta_n, \quad (5.11)$$

*where the product converges to a finite positive number.*

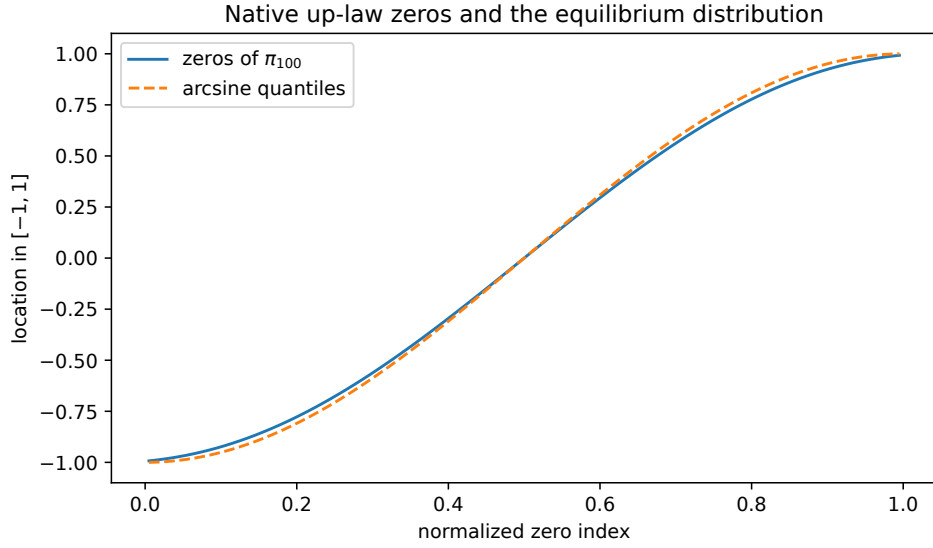


Figure 1: Zeros of  $\pi_{100}$ , computed as eigenvalues of the finite Jacobi matrix, against the arcsine quantiles. This is a numerical illustration of [equation \(5.7\)](#), not an additional proof.

*Proof.* Equation [equation \(5.10\)](#) is Szegő’s strong norm theorem in monic normalization; the constant is fixed by checking the model weight  $w \equiv 1$ , for which  $4^n H_n \rightarrow \pi$ . Since  $H_0 = 1$  and  $H_n = \prod_{j=1}^n \beta_j$ , one has the exact finite identity  $4^n H_n = \prod_{j=1}^n 4\beta_j$ . Passing to the limit proves [equation \(5.11\)](#).  $\square$

The partial product through  $n = 200$  is

$$\prod_{n=1}^{200} 4\beta_n = 0.00213943931416523273491231 \dots \quad (5.12)$$

This number is *not* presented as an accurate decimal evaluation of  $\mathcal{C}_F$ : the convergence is slow enough that the uncomputed tail is material. Its role is to certify positivity and provide data for future extrapolation.

**Corollary 5.5** (Hankel determinant asymptotic). As  $N \rightarrow \infty$ ,

$$\Delta_N = \mathcal{C}_F^N 2^{-N(N-1)} \exp(o(N)). \quad (5.13)$$

In particular,

$$\Delta_N^{1/N^2} \longrightarrow \frac{1}{2}. \quad (5.14)$$

*Proof.* The Gram identity gives  $\Delta_N = \prod_{n=0}^{N-1} H_n$ . From [equation \(5.10\)](#),  $\log H_n = \log \mathcal{C}_F - n \log 4 + o(1)$ . Summation gives

$$\begin{aligned} \log \Delta_N &= N \log \mathcal{C}_F - \log 4 \sum_{n=0}^{N-1} n + o(N) \\ &= N \log \mathcal{C}_F - N(N-1) \log 2 + o(N), \end{aligned}$$

which is [equation \(5.13\)](#). Divide by  $N^2$  and exponentiate to obtain [equation \(5.14\)](#).  $\square$

The capacity-scale limit  $1/2$  is universal for regular measures on  $[-1, 1]$ ; all Fabius-specific information surviving at the next exponential scale is compressed into  $\mathcal{C}_F$ .

## 6 Christoffel reconstruction from rational moments

Christoffel functions turn the moment matrix into a local density estimator. For  $N \geq 1$ , let  $\mathcal{P}_{N-1}$  be the polynomials of degree below  $N$  and define

$$\lambda_N(x) = \min \left\{ \int_{-1}^1 |P(t)|^2 w(t) dt : P \in \mathcal{P}_{N-1}, P(x) = 1 \right\}. \quad (6.1)$$

If  $p_n = \pi_n / \sqrt{H_n}$ , the Christoffel–Darboux kernel and Christoffel function are

$$K_N(x, y) = \sum_{n=0}^{N-1} p_n(x) p_n(y), \quad \lambda_N(x) = \frac{1}{K_N(x, x)}. \quad (6.2)$$

### 6.1 Moment-matrix formula and exact arithmetic

Put

$$M_N = [m_{j+k}]_{j,k=0}^{N-1}, \quad v_N(x) = (1, x, \dots, x^{N-1})^\top. \quad (6.3)$$

**Proposition 6.1** (Inverse-Hankel formula). *For every real  $x$ ,*

$$K_N(x, x) = v_N(x)^\top M_N^{-1} v_N(x), \quad \lambda_N(x) = \frac{1}{v_N(x)^\top M_N^{-1} v_N(x)}. \quad (6.4)$$

*In particular,  $\lambda_N(x) \in \mathbb{Q}$  whenever  $x \in \mathbb{Q}$ .*

*Proof.* For a coefficient vector  $c$ , the squared norm of  $P(t) = c^\top v_N(t)$  is  $c^\top M_N c$ . Minimize this positive definite quadratic form subject to  $c^\top v_N(x) = 1$ . A Lagrange multiplier gives  $c = M_N^{-1} v_N(x) / (v_N(x)^\top M_N^{-1} v_N(x))$ , and substitution yields [equation \(6.4\)](#). The entries of  $M_N$  and its inverse are rational, so rational  $x$  gives a rational value.  $\square$

The proposition makes every Christoffel value at a dyadic point an exact rational expression in the Bell–Bernoulli moment data. It does not sample  $w$  at that point; it infers local density from the first  $2N - 2$  moments.

### 6.2 Bulk reconstruction theorem

For positive continuous weights, the Máté–Nevai–Totik asymptotic is [\[8, 9, 10\]](#)

$$N \lambda_N(x) \longrightarrow \pi w(x) \sqrt{1 - x^2}, \quad -1 < x < 1. \quad (6.5)$$

The convergence is locally uniform wherever the weight is continuous and positive. Those hypotheses hold on every compact subinterval of  $(-1, 1)$ .

**Theorem 6.2** (Pointwise reconstruction of Rvachev’s bump). *For every  $x \in (-1, 1)$ ,*

$$w(x) = \text{up}(x) = \lim_{N \rightarrow \infty} \frac{N \lambda_N(x)}{\pi \sqrt{1 - x^2}}. \quad (6.6)$$

*The convergence is locally uniform in  $x$ .*

This is a reconstruction of a nowhere-analytic  $C^\infty$  function from finite rational moment matrices. At a rational  $x$ , every approximant is a rational number divided only by the displayed factor  $\pi \sqrt{1 - x^2}$ . At the symmetry point the algebraic factor disappears.



**Corollary 6.3** (A canonical rational limit for  $\pi$ ). *Since  $w(0) = 1$ ,*

$$\boxed{N\lambda_N(0) \longrightarrow \pi.} \quad (6.7)$$

*Every term  $N\lambda_N(0)$  is rational.*

The first paired values are

$$\begin{aligned} \lambda_1(0) &= \lambda_2(0) = 1, \\ \lambda_3(0) &= \lambda_4(0) = \frac{32}{57}, \\ \lambda_5(0) &= \lambda_6(0) = \frac{26727424}{67149435}. \end{aligned} \quad (6.8)$$

Consequently,

$$2\lambda_2(0) = 2, \quad 4\lambda_4(0) = \frac{128}{57} = 2.245614\dots, \quad 6\lambda_6(0) = \frac{53454848}{22383145} = 2.388174\dots \quad (6.9)$$

Parity explains the repetitions:  $p_{2j+1}(0) = 0$ , hence adding an odd polynomial to the kernel does not change its central value.

### 6.3 A moment reconstruction of the Fabius function itself

For  $0 < x < 1$ , the translation identity  $F(x) = w(x - 1)$  turns [equation \(6.6\)](#) into

$$\boxed{F(x) = \lim_{N \rightarrow \infty} \frac{N\lambda_N(x - 1)}{\pi\sqrt{x(2 - x)}}.} \quad (6.10)$$

Thus the full bounded Fabius function can be recovered from its even moments, without direct recursion on dyadic intervals. Formula [equation \(6.10\)](#) also proposes a new approximation scheme for the inverse: solve the finite algebraic equation

$$\frac{N\lambda_N(x - 1)}{\pi\sqrt{x(2 - x)}} = y \quad (6.11)$$

for  $x \in (0, 1)$ . The left side is an explicit rational function times one square root. Bulk convergence controls this procedure away from  $x = 0, 1$ ; the endpoint Lambert- $W$  regime requires a separate double-scaling analysis.

*Remark 6.4* (Why the endpoint is a genuine second problem). The factor  $\sqrt{1 - x^2}$  in [equation \(6.5\)](#) belongs to the universal equilibrium density. Near  $x = 1$ , it competes with the much smaller logarithmic-square density  $w(x)$ . Classical bulk asymptotics are therefore not uniform on scales where  $1 - x$  shrinks with  $N$ . Matching the Christoffel kernel to the repository's Lambert- $W$  endpoint saddle is one of the most promising open directions in [section 13](#).

## 7 An alternating rational Jacobi product for $\pi$

The center  $x = 0$  is special for an even measure. Here the strong Szegő interior asymptotic becomes an exact product statement about alternating Jacobi coefficients.

### 7.1 Strong center asymptotics

The strong interior Szegő formula for an even weight gives

$$p_{2M}(0)^2 \longrightarrow \frac{2}{\pi w(0)}. \quad (7.1)$$

For the up-law  $w(0) = 1$ , so the limit is  $2/\pi$ .

**Lemma 7.1** (Exact central product). *For every  $M \geq 1$ ,*

$$\pi_{2M}(0) = (-1)^M \prod_{j=1}^M \beta_{2j-1}, \quad (7.2)$$

and

$$p_{2M}(0)^2 = \prod_{j=1}^M \frac{\beta_{2j-1}}{\beta_{2j}}. \quad (7.3)$$

*Proof.* Set  $x = 0$  in [equation \(4.4\)](#). Odd polynomials vanish and  $\pi_{2M}(0) = -\beta_{2M-1}\pi_{2M-2}(0)$ , which iterates to [equation \(7.2\)](#). Also  $H_{2M} = \prod_{n=1}^{2M} \beta_n$ . Divide the square of [equation \(7.2\)](#) by  $H_{2M}$  to obtain [equation \(7.3\)](#).  $\square$

**Theorem 7.2** (Fabius–Jacobi product for  $\pi$ ). *The rational partial products*

$$\Pi_M = 2 \prod_{j=1}^M \frac{\beta_{2j}}{\beta_{2j-1}} \quad (7.4)$$

converge to  $\pi$ . *Equivalently,*

$$\pi = 2 \prod_{j=1}^{\infty} \frac{\beta_{2j}}{\beta_{2j-1}}. \quad (7.5)$$

*Proof.* By [lemma 7.1](#),  $\Pi_M = 2/p_{2M}(0)^2$ . Apply [equation \(7.1\)](#) with  $w(0) = 1$ .  $\square$

The formula is not a generic tautology. For a general even Szegő weight, identical reasoning gives

$$\pi w(0) = 2 \prod_{j=1}^{\infty} \frac{\beta_{2j}}{\beta_{2j-1}}. \quad (7.6)$$

The up-law is distinguished by the exact normalization  $w(0) = 1$ , so the spectral product isolates  $\pi$  itself.

### 7.2 Hankel-determinant form

Substitution of  $\beta_n = \Delta_{n+1}\Delta_{n-1}/\Delta_n^2$  into [equation \(7.5\)](#) gives a formula involving only exact moment Hankel determinants:

$$\pi = 2 \prod_{j=1}^{\infty} \frac{\Delta_{2j+1}\Delta_{2j-1}^3}{\Delta_{2j}^3\Delta_{2j-2}}. \quad (7.7)$$

Every determinant in [equation \(7.7\)](#) is a rational expression in the Fabius moments, which in turn are generated by Bernoulli/Bell data or the triangular recurrence. This supplies a direct arithmetic chain

$$\text{Bernoulli cumulants} \longrightarrow \text{Bell moments} \longrightarrow \text{Hankel determinants} \longrightarrow \pi. \quad (7.8)$$

### 7.3 Exact harmonic mean of the product approximants

Set  $\Pi_0 = 2$ . Since  $p_{2m+1}(0) = 0$ ,

$$K_{2M}(0, 0) = \sum_{m=0}^{M-1} p_{2m}(0)^2 = 2 \sum_{m=0}^{M-1} \frac{1}{\Pi_m}. \quad (7.9)$$

**Theorem 7.3** (Christoffel–product harmonic mean). *For every  $M \geq 1$ ,*

$$\boxed{2M\lambda_{2M}(0) = \frac{M}{\sum_{m=0}^{M-1} \Pi_m^{-1}}}. \quad (7.10)$$

*Thus the Christoffel approximant is exactly the harmonic mean of  $\Pi_0, \dots, \Pi_{M-1}$ .*

*Proof.* Invert [equation \(7.9\)](#) and multiply by  $2M$ . □

This identity explains why the direct products converge much faster in the experiments than  $2M\lambda_{2M}(0)$ : the latter averages the entire history, including the crude early values.

**Corollary 7.4** (Conditional first correction). *Suppose*

$$\mathcal{S}_F = \sum_{m=0}^{\infty} \left( \frac{1}{\Pi_m} - \frac{1}{\pi} \right) \quad (7.11)$$

*converges. Then*

$$2M\lambda_{2M}(0) = \pi - \frac{\pi^2 \mathcal{S}_F}{M} + o(M^{-1}). \quad (7.12)$$

*Proof.* The denominator in [equation \(7.10\)](#) is  $M/\pi + \mathcal{S}_F + o(1)$ . Expand its reciprocal. □

The convergence assumed here follows, for example, from the product-error conjecture in [conjecture 11.2](#). The data suggest  $M(\pi - 2M\lambda_{2M}(0))$  approaches a finite positive constant; at  $M = 100$  it is approximately 5.8867251451.

### 7.4 Initial rational products

The first two exact products and several high-precision values are

$$\begin{aligned} \Pi_1 &= \frac{64}{25} = 2.56, \\ \Pi_2 &= \frac{53454848}{19541211} = 2.7354931073616676\dots, \\ \Pi_3 &= 2.8320987418684373\dots, & \Pi_4 &= 2.9033763206778567\dots, \\ \Pi_5 &= 2.9457847614454995\dots, & \Pi_{10} &= 3.0478239630996334\dots, \\ \Pi_{20} &= 3.1010805882480384\dots, & \Pi_{50} &= 3.1298973222596123\dots, \\ \Pi_{100} &= 3.1373662375494674\dots \end{aligned} \quad (7.13)$$

The fact that these values increase and remain below  $\pi$  is empirical at present; convergence itself is proved by [theorem 7.2](#).

## 8 Gauss quadrature and Padé approximation of the Cauchy transform

The repository already proves the Laurent expansion and the affine-difference renormalization of the up-law Cauchy transform. Native orthogonal polynomials supply the missing continued-fraction and optimal-quadrature layer.

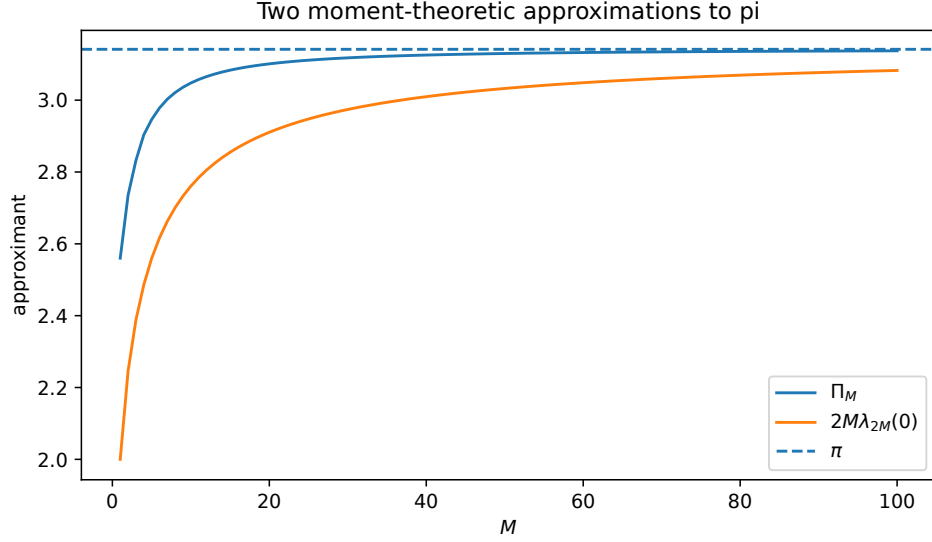


Figure 2: The direct rational products  $\Pi_M$  and the exact Christoffel harmonic means  $2M\lambda_{2M}(0)$ . Both converge to  $\pi$ ; monotone approach from below is conjectural.

### 8.1 Exact Gaussian rules

Let  $x_{N,1} < \dots < x_{N,N}$  be the zeros of  $\pi_N$ . They are simple and belong to  $(-1, 1)$ . Define

$$\omega_{N,j} = \lambda_N(x_{N,j}) = \frac{1}{K_N(x_{N,j}, x_{N,j})} > 0. \quad (8.1)$$

**Theorem 8.1** (Native up-law Gauss quadrature). *For every polynomial  $P$  of degree at most  $2N - 1$ ,*

$$\int_{-1}^1 P(x)w(x) dx = \sum_{j=1}^N \omega_{N,j} P(x_{N,j}). \quad (8.2)$$

*The nodes are algebraic, the weights are positive algebraic numbers, and all symmetric sums required to verify equation (8.2) belong to  $\mathbb{Q}$ .*

The first rules are especially simple:

$$\begin{aligned} N = 1 : \quad & x_{1,1} = 0, & \omega_{1,1} &= 1; \\ N = 2 : \quad & x_{2,1}, x_{2,2} = \mp \frac{1}{3}, & \omega_{2,1} = \omega_{2,2} &= \frac{1}{2}; \end{aligned} \quad (8.3)$$

$$N = 3 : \quad x_{3,1}, x_{3,2}, x_{3,3} = -\sqrt{\frac{19}{75}}, 0, \sqrt{\frac{19}{75}}, \quad \omega_{3,2} = \frac{32}{57}, \quad \omega_{3,1} = \omega_{3,3} = \frac{25}{114}. \quad (8.4)$$

Thus the two-node rule integrates exactly through degree three and the three-node rule through degree five against the non-elementary density  $w$ . These formulae are useful for integrals involving  $F$  because  $F(x) = w(x-1)$  on  $[0, 1]$ .

### 8.2 The Jacobi continued fraction

For  $z \in \mathbb{C} \setminus [-1, 1]$ , define

$$\mathcal{R}(z) = \int_{-1}^1 \frac{w(x)}{z-x} dx. \quad (8.5)$$

At infinity,

$$\mathcal{R}(z) = \sum_{k=0}^{\infty} \frac{m_k}{z^{k+1}}, \quad |z| > 1, \quad (8.6)$$

with the exact finite remainders already formalized in the repository. Stieltjes–Markov theory [5, 12] gives the  $J$ -fraction

$$\mathcal{R}(z) = \cfrac{1}{z - \cfrac{\beta_1}{z - \cfrac{\beta_2}{z - \cfrac{\beta_3}{\ddots}}}}. \quad (8.7)$$

The fraction converges locally uniformly off  $[-1, 1]$ .

Let

$$Q_{N-1}(z) = \int_{-1}^1 \frac{\pi_N(z) - \pi_N(x)}{z - x} w(x) dx. \quad (8.8)$$

Then  $Q_{N-1} \in \mathbb{Q}[z]$  has degree  $N - 1$  and the  $N$ th convergent of equation (8.7) is  $Q_{N-1}/\pi_N$ .

**Theorem 8.2** (Gauss–Padé identity and exact error). *The rational function  $Q_{N-1}/\pi_N$  is the  $[N - 1/N]$  Padé approximant of equation (8.6) at infinity. Moreover,*

$$\mathcal{R}(z) - \frac{Q_{N-1}(z)}{\pi_N(z)} = \frac{1}{\pi_N(z)^2} \int_{-1}^1 \frac{\pi_N(x)^2}{z - x} w(x) dx. \quad (8.9)$$

For real  $z > 1$  this error is strictly positive and satisfies

$$\frac{H_N}{(z + 1)\pi_N(z)^2} \leq \mathcal{R}(z) - \frac{Q_{N-1}(z)}{\pi_N(z)} \leq \frac{H_N}{(z - 1)\pi_N(z)^2}. \quad (8.10)$$

*Proof.* From equation (8.8),

$$\pi_N(z)\mathcal{R}(z) - Q_{N-1}(z) = \int \frac{\pi_N(x)}{z - x} d\mu(x).$$

The polynomial  $(\pi_N(z) - \pi_N(x))/(z - x)$  has degree  $N - 1$  in  $x$ , so it is orthogonal to  $\pi_N(x)$ . Therefore

$$\pi_N(z) \int \frac{\pi_N(x)}{z - x} d\mu(x) = \int \frac{\pi_N(x)^2}{z - x} d\mu(x),$$

which proves equation (8.9). Expansion at infinity and orthogonality show that the first  $2N$  Laurent coefficients agree, giving the Padé order. For  $z > 1$ ,  $(z + 1)^{-1} \leq (z - x)^{-1} \leq (z - 1)^{-1}$  on the support; integration gives equation (8.10).  $\square$

Partial fractions recover the Gaussian rule:

$$\frac{Q_{N-1}(z)}{\pi_N(z)} = \sum_{j=1}^N \frac{\omega_{N,j}}{z - x_{N,j}}. \quad (8.11)$$

Thus Gaussian quadrature and Padé approximation are two faces of the same finite Jacobi truncation.

The first nontrivial convergents are

$$\begin{aligned} \mathcal{R}_1(z) &= \frac{1}{z}, \\ \mathcal{R}_2(z) &= \frac{z}{z^2 - 1/9}, \\ \mathcal{R}_3(z) &= \frac{z^2 - 32/225}{z(z^2 - 19/75)}. \end{aligned} \quad (8.12)$$

For rational  $z > 1$ , each value is rational and is a certified lower bound for  $\mathcal{R}(z)$ ; [equation \(8.10\)](#) provides a fully explicit rational error interval.

### 8.3 Coupling to dyadic renormalization

The repository proves

$$\mathcal{R}'(z) = 2(\mathcal{R}(2z + 1) - \mathcal{R}(2z - 1)), \quad z \in \mathbb{C} \setminus [-1, 1]. \quad (8.13)$$

Equations [equations \(8.7\)](#) and [\(8.13\)](#) are complementary: the first is spectral and nonlinear in the sequence  $\beta_n$ , while the second is dyadic and linear in the function  $\mathcal{R}$ . Substitution of the finite convergents  $\mathcal{R}_N$  into [equation \(8.13\)](#) creates explicit rational residuals

$$\mathfrak{E}_N(z) = \mathcal{R}'_N(z) - 2(\mathcal{R}_N(2z + 1) - \mathcal{R}_N(2z - 1)). \quad (8.14)$$

The exact error formula bounds  $\mathfrak{E}_N$  on compact subsets of the slit plane. Studying its poles and signs may convert the dyadic functional law into new inequalities for the Jacobi coefficients; this program is stated more precisely in [section 13](#).

## 9 The Legendre–Gaunt determinant bridge

The preceding sections used only the native moments. We now connect them to the repository's exact Fourier–Legendre expansion, thereby closing a finite algebraic circuit between the classical and native orthogonal systems.

### 9.1 Exact Legendre data

Let  $P_n$  be the standard Legendre polynomial,  $P_n(1) = 1$ . The repository proves an absolutely and uniformly convergent expansion

$$w(x) = \sum_{r=0}^{\infty} c_r P_{2r}(x), \quad c_r = \frac{4r+1}{2} \int_{-1}^1 w(x) P_{2r}(x) dx, \quad (9.1)$$

with  $c_r \in \mathbb{Q}$ . Only even degrees occur because  $w$  is even.

For  $N \geq 1$ , form the Gram matrix of the first  $N$  Legendre polynomials in the  $up$  inner product:

$$G_N = \left[ \int_{-1}^1 P_j(x) P_k(x) w(x) dx \right]_{j,k=0}^{N-1}. \quad (9.2)$$

For  $N = 0$ , set  $G_0$  to be the empty  $0 \times 0$  Gram matrix. This standard convention extends the preceding  $N \geq 1$  definition and gives  $\det G_0 = 1$ .

### 9.2 Finite Gaunt expansion

The Gaunt integral is

$$\int_{-1}^1 P_j(x) P_k(x) P_\ell(x) dx = 2 \begin{pmatrix} j & k & \ell \\ 0 & 0 & 0 \end{pmatrix}^2. \quad (9.3)$$

It vanishes unless the triangle inequalities hold and  $j + k + \ell$  is even. Consequently the apparently infinite substitution of [equation \(9.1\)](#) into [equation \(9.2\)](#) truncates exactly.

**Theorem 9.1** (Finite rational Gaunt matrix). *For  $0 \leq j, k < N$ ,*

$$(G_N)_{jk} = 2 \sum_{r \geq 0} c_r \binom{j \quad k \quad 2r}{0 \quad 0 \quad 0}^2, \quad (9.4)$$

where only  $|j - k| \leq 2r \leq j + k$  contribute. Thus every entry of  $G_N$  is a finite rational sum.

For completeness, if  $j + k + \ell = 2s$ , the squared symbol is

$$\binom{j \quad k \quad \ell}{0 \quad 0 \quad 0}^2 = \frac{s!^2}{(s-j)!^2 (s-k)!^2 (s-\ell)!^2} \frac{(2s-2j)!(2s-2k)!(2s-2\ell)!}{(2s+1)!}, \quad (9.5)$$

otherwise it is zero. Formula [equation \(9.5\)](#) makes the rationality in [equation \(9.4\)](#) completely explicit.

### 9.3 Determinants recover the native recurrence

The leading coefficient of  $P_n$  is

$$L_n = 2^{-n} \binom{2n}{n}. \quad (9.6)$$

Let  $D_N = \det G_N$ , so in particular the empty-determinant convention gives  $D_0 = 1$ . The triangular change of basis from monomials to  $P_0, \dots, P_{N-1}$  has determinant  $\prod_{j=0}^{N-1} L_j$ . Hence:

**Theorem 9.2** (Legendre–Hankel determinant identity). *For every  $N \geq 0$ ,*

$$D_N = \left( \prod_{j=0}^{N-1} L_j^2 \right) \Delta_N. \quad (9.7)$$

Consequently, for  $n \geq 1$ ,

$$\beta_n = \left( \frac{n}{2n-1} \right)^2 \frac{D_{n+1} D_{n-1}}{D_n^2}. \quad (9.8)$$

*Proof.* Writing the Legendre coefficient vector as a triangular matrix times the monomial coefficient vector gives  $G_N = A_N M_N A_N^T$ , with  $\det A_N = \prod_{j < N} L_j$ . This proves [equation \(9.7\)](#). Substitute it into  $\beta_n = \Delta_{n+1} \Delta_{n-1} / \Delta_n^2$ . The products telescope to  $(L_{n-1} / L_n)^2$ , and  $L_n / L_{n-1} = (2n-1)/n$ .  $\square$

**Remark 9.3** (Exact Lean crosswalk and indexing). Equation (9.7) is the specialization `upLegendreGramDet_eq_prod_leadingCoeff_sq_mul_hankelDet`. The theorem `gramStieltjesJacobiSubdiagonal_upMoment_eq_upLegendreGramDet_ratio` and its rational-cast companion `rvachevJacobiSubdiagonalRat_cast_eq_upLegendreGramDet_ratio` use index  $k \geq 0$  for the conventional coefficient  $\beta_{k+1}$ ; replacing  $k$  by  $n-1$  gives exactly (9.8). Their proof is the displayed  $G = C^T H C$  determinant argument and does not pass through the still-open Gaunt/Wigner entry expansion. The determinant identity,  $D_0 = 1$ , the leading-coefficient quotient, and the real Gram cross-ratio hold for every `BoundedFabius`; entry-as-integral, strict positivity, and the rational-cast bridge require `IsFabius`. Without Hankel nonvanishing, the generic and real Gram cross-ratios are unconditional only through totalized division: a zero middle determinant makes both sides zero and does not supply a nonsingular Jacobi recurrence.

A low-degree check is instructive. From the moments,

$$G_3 = \begin{pmatrix} 1 & 0 & -1/3 \\ 0 & 1/9 & 0 \\ -1/3 & 0 & 11/75 \end{pmatrix}, \quad D_3 = \frac{8}{2025}. \quad (9.9)$$

Since  $D_1 = 1$  and  $D_2 = 1/9$ , [equation \(9.8\)](#) gives

$$\beta_2 = \left(\frac{2}{3}\right)^2 \frac{(8/2025) \cdot 1}{(1/9)^2} = \frac{32}{225}, \quad (9.10)$$

matching [equation \(4.7\)](#).

#### 9.4 Christoffel functions directly in the Legendre basis

Let

$$\mathbf{P}_N(x) = (P_0(x), \dots, P_{N-1}(x))^T. \quad (9.11)$$

Basis invariance of the reproducing kernel gives

$$K_N(x, x) = \mathbf{P}_N(x)^T G_N^{-1} \mathbf{P}_N(x). \quad (9.12)$$

Combining [equations \(9.4\)](#) and [\(9.12\)](#) with [equation \(6.6\)](#) yields a representation of  $w(x)$  using only finite rational combinations of the existing exact Legendre coefficients  $c_r$ .

This closes the promised circuit:

$$\begin{aligned} &\text{exact dyadic values of } F \longrightarrow c_r \longrightarrow G_N \text{ by Gaunt sums} \longrightarrow D_N \longrightarrow \beta_n \\ &\longrightarrow \begin{cases} \Pi_M \rightarrow \pi, \\ \lambda_N(x) \rightarrow \pi w(x) \sqrt{1-x^2}/N, \end{cases} \longrightarrow w = \text{up} \longrightarrow F. \end{aligned} \quad (9.13)$$

Unlike the shifted-up self-reconstruction in the recent Legendre reports, all intermediate matrices in [equation \(9.13\)](#) are finite, positive definite, and basis invariant.

## 10 Numerical experiment and exact verification

The archive contains the standalone program `fabius_frontier_experiments.py`. It uses no sampled values of  $F$  or  $w$ . Its only analytic input is the exact moment recurrence [equation \(2.11\)](#); all later quantities are generated by the native three-term recurrence.

### 10.1 Two arithmetic modes

The program performs two independent calculations.

1. **Exact mode.** Python's `Fraction` type generates the moments, polynomials, norms,  $\beta_n$ ,  $\Pi_M$ , and central Christoffel approximants through degree 24. Exact assertions check [equation \(4.7\)](#) through  $\pi_4$  and  $\beta_3$ .
2. **High-precision mode.** The same recurrence is evaluated with `mpmath` at 4,800 decimal digits through  $\beta_{200}$ . Every coefficient through degree 24 is compared with the independent exact calculation. Positivity of every computed squared norm is also checked before any table or figure is written.

The identity

$$H_n = \int x^n \pi_n(x) w(x) dx \quad (10.1)$$

reduces the recurrence to  $\mathcal{O}(N^2)$  coefficient operations. Thousands of digits are nevertheless necessary because monomial coefficients and moments cancel severely at high degree. A failed longer trial was discarded in full; none of its partial files or values are used in this report.



Table 2: Selected high-precision Jacobi coefficients. The last column tests the scaling proposed in [conjecture 11.2](#).

| $n$ | $\beta_n$            | $\frac{n^2(1/4 - \beta_n)}{(\log_2 n)^2}$ |
|-----|----------------------|-------------------------------------------|
| 10  | 0.218406233337115543 | 0.286299738281                            |
| 20  | 0.234873012373991668 | 0.323934622747                            |
| 40  | 0.243489365492140411 | 0.367794553321                            |
| 60  | 0.246188400107704611 | 0.393271027136                            |
| 80  | 0.247434883352791960 | 0.410759852630                            |
| 100 | 0.248128725130487816 | 0.423932916190                            |
| 150 | 0.248962552551866080 | 0.446697951249                            |
| 200 | 0.249325249690015095 | 0.461932021189                            |

## 10.2 Observed recurrence coefficients

The validated values include: All 200 computed coefficients satisfy

$$0 < \beta_1 < \beta_2 < \cdots < \beta_{200} < \frac{1}{4}. \quad (10.2)$$

Only the endpoint limit  $\beta_n \rightarrow 1/4$  is proved; the strict inequalities in [equation \(10.2\)](#) are evidence for [conjecture 11.1](#).

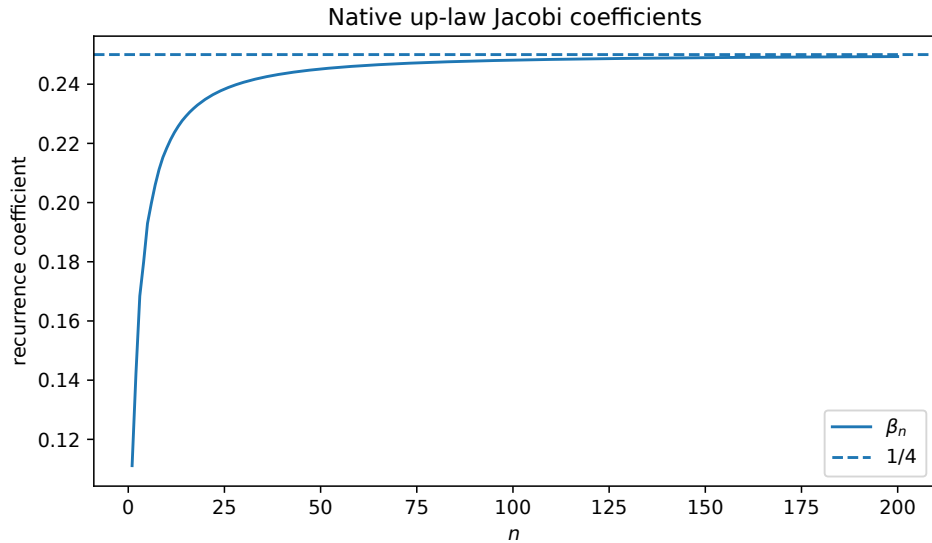


Figure 3: The computed native recurrence coefficients and the proved limit  $1/4$ .

## 10.3 Observed product errors

For  $\Pi_M = 2 \prod_{j \leq M} \beta_{2j} / \beta_{2j-1}$ , the diagnostic

$$E_M = \frac{M^2(\pi - \Pi_M)}{(\log_2 M)^2} \quad (10.3)$$

has the following values:

|       |          |          |          |          |          |
|-------|----------|----------|----------|----------|----------|
| $M$   | 10       | 20       | 50       | 75       | 100      |
| $E_M$ | 0.849723 | 0.867540 | 0.917911 | 0.941369 | 0.957485 |

(10.4)

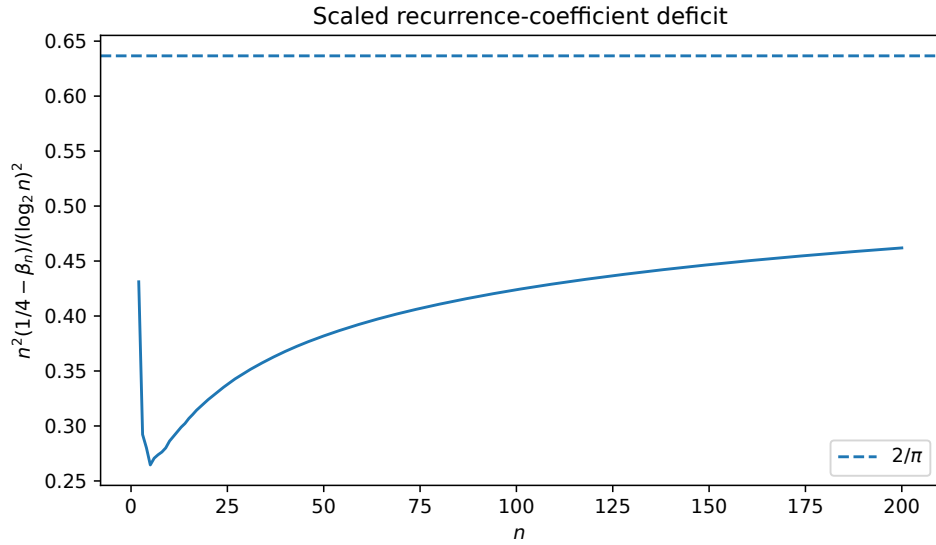


Figure 4: The scaled deficit  $n^2(1/4 - \beta_n)/(\log_2 n)^2$ . The horizontal line is the conjectural limit  $2/\pi$ , not a fitted value.

The drift toward 1 is substantially clearer than the corresponding drift toward  $2/\pi$  in [table 2](#); the alternating product cancels part of the slowly varying lower-order structure.

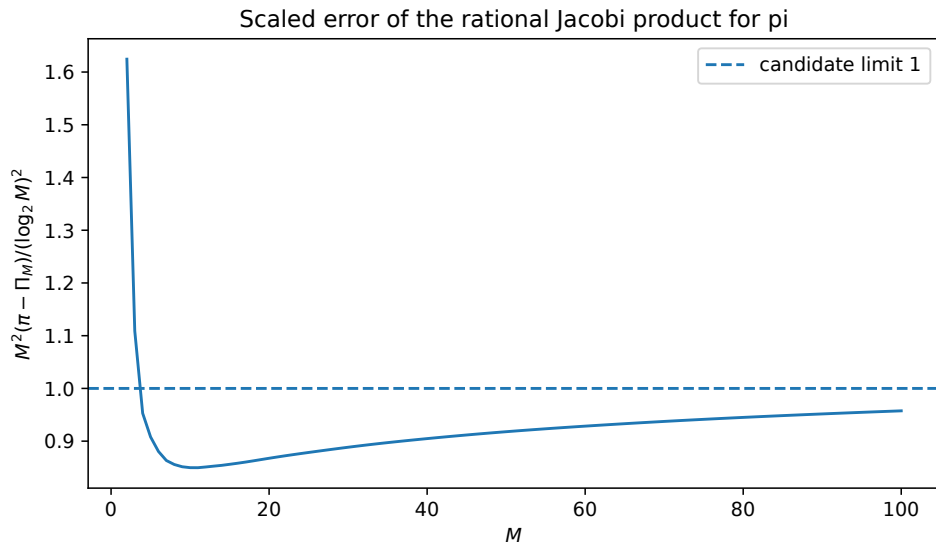


Figure 5: Scaled error of the rational product. The candidate limit 1 is part of [conjecture 11.2](#).

## 10.4 Reproducibility and data products

The command used for the archived run is

```
python fabius_frontier_experiments.py \
  --degree 200 --dps 4800 --output .
```

It writes:

- `data/exact_low_degree.csv`, containing exact fractions;

- `data/jacobi_coefficients.csv`, containing  $\beta_1, \dots, \beta_{200}$  and scaled deficits;
- `data/pi_product_approximants.csv`, containing the products, Christoffel harmonic means, and errors;
- the five PDF figures reproduced in this report; and
- `experiment_run.log`, the compact diagnostic transcript.

The script deliberately exposes its precision and degree as command-line parameters. Increasing the degree without increasing precision can silently destroy positivity, so the program emits a warning below a conservative precision threshold and checks exact low-degree overlap on every run.

## 11 Conjectures suggested by the spectral data

All statements in this section are open within the present report. They are ordered from structurally plausible to increasingly speculative.

### 11.1 Monotonicity and one-sided rational approximants

**Conjecture 11.1** (Strict Nevai approach). *For every  $n \geq 1$ ,*

$$0 < \beta_n < \beta_{n+1} < \frac{1}{4}. \quad (11.1)$$

If true, each ratio  $\beta_{2j}/\beta_{2j-1}$  exceeds one. Together with [theorem 7.2](#), this would prove

$$2 < \Pi_1 < \Pi_2 < \dots < \pi, \quad (11.2)$$

turning [equation \(7.4\)](#) into a canonical sequence of exact rational lower bounds for  $\pi$ . The harmonic-mean identity would then imply that  $2M\lambda_{2M}(0)$  is also increasing and lies below  $\Pi_{M-1} < \pi$ . A proof may require a new comparison theorem for recurrence coefficients under successive uniform convolution; generic log-concavity alone does not imply [equation \(11.1\)](#).

### 11.2 A logarithmic-square correction at the Nevai edge

**Conjecture 11.2** (Matched Jacobi and product rates). *As  $n, M \rightarrow \infty$ ,*

$$\frac{1}{4} - \beta_n \sim \frac{2}{\pi} \frac{(\log_2 n)^2}{n^2}, \quad (11.3)$$

$$\pi - \Pi_M \sim \frac{(\log_2 M)^2}{M^2}. \quad (11.4)$$

The appearance of the same logarithmic square as in [equation \(2.13\)](#) is natural: recurrence coefficients at order  $n$  resolve endpoint structure on a scale of order  $n^{-2}$ . The numerical constant  $2/\pi$  is less obvious. It is supported both by [figure 4](#) and by the internal consistency calculation below.

**Proposition 11.3** (Conditional matching of the two constants). *Suppose  $d_n = 1/4 - \beta_n$  has a sufficiently smooth regularly varying asymptotic*

$$d_n \sim A \frac{(\log_2 n)^2}{n^2}, \quad d_{n-1} - d_n \sim 2A \frac{(\log_2 n)^2}{n^3}. \quad (11.5)$$

Then

$$\pi - \Pi_M \sim \frac{\pi A}{2} \frac{(\log_2 M)^2}{M^2}. \quad (11.6)$$

Thus the candidate coefficient 1 in [equation \(11.4\)](#) forces  $A = 2/\pi$ .

*Proof.* Since  $d_n \rightarrow 0$ ,  $\log \beta_n = -\log 4 - 4d_n + \mathcal{O}(d_n^2)$ . Therefore

$$\begin{aligned} \log \frac{\pi}{\Pi_M} &= \sum_{j>M} (\log \beta_{2j} - \log \beta_{2j-1}) \\ &= 4 \sum_{j>M} (d_{2j-1} - d_{2j}) + o\left(\frac{(\log M)^2}{M^2}\right). \end{aligned}$$

Using [equation \(11.5\)](#) at  $n = 2j$  and  $\sum_{j>M} j^{-3} \sim (2M^2)^{-1}$  gives

$$\log \frac{\pi}{\Pi_M} \sim \frac{A}{2} \frac{(\log_2 M)^2}{M^2}.$$

Because this tends to zero, exponentiation gives [equation \(11.6\)](#).  $\square$

A proof of [equation \(11.3\)](#) likely lies beyond classical strong Szegő theory. The weight vanishes at both endpoints faster than every power, with a logarithmic-square exponent and dyadic lower-order oscillation. A Riemann–Hilbert or nonlinear steepest-descent analysis adapted to this “log-Gaussian hard edge” appears necessary.

### 11.3 Dyadic periodic modulation

The repository’s Mellin and Lambert analyses repeatedly produce periodic functions of  $\log_2$  of the scale. It is therefore reasonable to expect a smaller spectral echo.

**Conjecture 11.4** (Spectral dyadic ripple). *There exist bounded one-periodic functions  $A_1, A_0$  such that, with  $L = \log_2 n$ ,*

$$n^2 \left( \frac{1}{4} - \beta_n \right) = \frac{2}{\pi} L^2 + L A_1(\{L\}) + A_0(\{L\}) + o(1). \quad (11.7)$$

*At least one of  $A_1, A_0$  is nonconstant.*

The present range is too short to distinguish a genuine periodic ripple from a long transient. Testing [equation \(11.7\)](#) requires a better conditioned recurrence algorithm, preferably based on modified moments in a Chebyshev basis rather than monomials.

### 11.4 Strict log-concavity and strict Turán inequalities

**Conjecture 11.5** (Strict interior shape). *The function  $\log w$  is strictly concave on  $(-1, 1)$ . Equivalently,  $\log F$  is strictly concave on  $(0, 1)$ . In particular,*

$$F(2x)^2 > 2F(x)F(4x), \quad 0 < x < \frac{1}{4}. \quad (11.8)$$

The conjecture should follow from equality-case analysis for Prékopa’s theorem applied to at least two nonproportional uniform summands, followed by a limit argument. A direct differential approach would amount to proving strict decrease of  $F(2x)/F(x)$ .

### 11.5 The central Christoffel shift

**Conjecture 11.6** (Finite Christoffel shift). *The series  $\mathcal{S}_F$  in [equation \(7.11\)](#) converges and is positive. Consequently*

$$M(\pi - 2M\lambda_{2M}(0)) \longrightarrow \pi^2 \mathcal{S}_F \in (0, \infty). \quad (11.9)$$

Conjecture [11.2](#) implies absolute convergence because  $\pi - \Pi_M = \mathcal{O}((\log M)^2/M^2)$ . The computed value at  $M = 100$  is 5.8867251451...; it should not yet be treated as a reliable estimate of the limit.

## 12 A geometric $q$ -uniform extension

The preceding structure is not isolated at the binary point. It extends to a natural geometric family that connects the repository's  $q$ -calculus to log-concavity and Jacobi products.

### 12.1 Definition, transform, and cumulants

Let  $0 < q < 1$ , let  $V_0, V_1, \dots$  be independent uniform random variables on  $[-1, 1]$ , and put

$$Y_q = \sum_{j=0}^{\infty} (1-q)q^j V_j. \quad (12.1)$$

The coefficients sum to one, so  $Y_q \in [-1, 1]$ . Denote its even density by  $w_q$ . At  $q = 1/2$ ,  $Y_q$  is exactly the up-law random variable  $Y$  and  $w_{1/2} = w$ .

The moment generating function is

$$M_q(z) = \prod_{j=0}^{\infty} \frac{\sinh((1-q)q^j z)}{(1-q)q^j z}. \quad (12.2)$$

Its even cumulants are

$$\kappa_{2r}(q) = \frac{2^{2r-1} B_{2r}}{r} \frac{(1-q)^{2r}}{1-q^{2r}}, \quad r \geq 1. \quad (12.3)$$

Thus rational  $q$  gives rational cumulants, moments, Hankel determinants, and Jacobi coefficients. Formula [equation \(12.3\)](#) specializes at  $q = 1/2$  to [equation \(2.9\)](#).

### 12.2 Log-concavity and the Szegő class

**Theorem 12.1** ( $q$ -uniform log-concavity). *For every  $0 < q < 1$ , the law of  $Y_q$  is log-concave and its continuous density  $w_q$  is log-concave on  $(-1, 1)$ .*

*Proof.* Every finite partial sum in [equation \(12.1\)](#) is a convolution of uniform log-concave measures. The partial sums converge almost surely and in distribution to  $Y_q$ . Closure under convolution and weak limits proves log-concavity of the limiting measure. Convolution with the first uniform summand gives a continuous density; its continuous representative is log-concave.  $\square$

A useful exact endpoint relation follows by conditioning on  $V_0$ . If  $T_q(t) = \Pr(Y_q \geq 1-t)$ , then for sufficiently small  $t > 0$ ,

$$w_q(1-t) = \frac{T_q(t/q)}{2(1-q)}. \quad (12.4)$$

A finite-coordinate small-box estimate, choosing  $N \asymp \log(1/t)/|\log q|$ , gives

$$-\log w_q(1-t) = \mathcal{O}(\log^2(1/t)). \quad (12.5)$$

Hence the proof of [theorem 5.1](#) applies verbatim and  $w_q$  belongs to the Szegő class. The sharper expected leading term is

$$-\log w_q(1-t) \sim \frac{\log^2(1/t)}{2|\log q|}, \quad (12.6)$$

but the exact normalization and periodic correction should be derived from a separate Mellin analysis; [equation \(12.6\)](#) is recorded as a research target rather than used below.

### 12.3 A continuum of Jacobi products

Let  $\beta_n(q)$  be the monic recurrence coefficients for  $w_q(x) dx$ . Rakhmanov and strong Szegő theory give

$$\beta_n(q) \longrightarrow \frac{1}{4} \quad (12.7)$$

and, by [equation \(7.6\)](#),

$$\pi w_q(0) = 2 \prod_{j=1}^{\infty} \frac{\beta_{2j}(q)}{\beta_{2j-1}(q)}. \quad (12.8)$$

At  $q = 1/2$ ,  $w_q(0) = 1$  and this is the new Fabius product [equation \(7.5\)](#).

At the degenerate endpoint  $q = 0$ , only  $V_0$  remains and  $w_0 \equiv 1/2$ . The recurrence coefficients become those of monic Legendre polynomials,

$$\beta_n(0) = \frac{n^2}{4n^2 - 1}. \quad (12.9)$$

Then [equation \(12.8\)](#) reduces to a Wallis-type product for  $\pi/2$ . Thus the family continuously deforms the classical Wallis/Legendre product into the Fabius product while preserving exact rationality whenever  $q \in \mathbb{Q}$ .

This interpolation suggests a two-parameter research program: determine the  $q$ -dependence of  $w_q(0)$  and of the first correction to  $\beta_n(q) \rightarrow 1/4$ , and study whether monotonicity in  $n$  or  $q$  survives.

## 13 Research program and formalization roadmap

The new results suggest several projects with sharply separated proof requirements. The first group is finite and algebraic; the last group needs new asymptotic analysis.

### 13.1 Prove monotonicity by a convolution flow

The measure  $\mu$  is obtained by repeatedly convolving with successively smaller uniform laws. Let  $\mu^{(N)}$  be the law of the first  $N$  summands and  $\beta_n^{(N)}$  its Jacobi coefficients. A productive route to [conjecture 11.1](#) is:

1. derive the response of finite Hankel determinants under convolution with a centered uniform law;
2. identify total-positivity or variation-diminishing inequalities that compare  $\beta_{n+1}^{(N)}$  and  $\beta_n^{(N)}$ ;
3. pass to the weak limit using moment convergence.

The heat flow drives Jacobi data through a Toda-type system. Uniform convolution is not a semigroup, but a discrete analogue may exist through the cumulant update

$$\kappa_{2r} \longmapsto \kappa_{2r} + \frac{2^{2r-1} B_{2r}}{r} a^{2r}. \quad (13.1)$$

Because the cumulant increment has alternating Bernoulli signs, naive coefficientwise positivity is unlikely; Hankel total positivity is the more plausible invariant.

### 13.2 Derive the log-square Jacobi correction

The principal analytic problem is to prove [equation \(11.3\)](#). A possible sequence of reductions is:

1. sharpen the endpoint formula to

$$\log w(1-t) = -\frac{\log^2(1/t)}{2\log 2} + A(\{\log_2(1/t)\}) \log(1/t) + B(\{\log_2(1/t)\}) + o(1), \quad (13.2)$$

with explicit periodic  $A, B$ ;

2. insert this weight into the Riemann–Hilbert problem for the native orthogonal polynomials;
3. build an endpoint parametrix for an exponential of a squared logarithm, rather than a Jacobi power  $t^\alpha$ ;
4. match it to the outer Szegő parametrix and read off the  $1/n^2$  correction to the recurrence coefficients.

The relevant endpoint scale should be found by balancing  $n\sqrt{t}$  against  $\log(1/t)$ , which is precisely where Lambert- $W$  enters. This makes the repository’s inverse-function asymptotics structurally relevant to the native spectral problem, not merely analogous to it.

### 13.3 Christoffel–Lambert double scaling

Formula [equation \(6.10\)](#) is a bulk statement. A uniform endpoint approximation would study

$$x = x_N \downarrow 0, \quad \frac{N\lambda_N(x_N - 1)}{\pi\sqrt{x_N(2 - x_N)}} \quad (13.3)$$

on scales where the right side is exponentially small. Three regimes should be separated:

1.  $x_N$  fixed: ordinary bulk asymptotics;
2.  $x_N \rightarrow 0$  slowly: a matching region controlled by the Szegő function;
3. the critical edge: a new kernel depending on a Lambert- $W$  scaled coordinate and possibly on the fractional part of  $\log_2(1/x_N)$ .

A successful analysis would give moment-based approximants to  $F^{-1}(y)$  with rigorous error bounds and would identify the analogue of the Bessel kernel for a logarithmic-square hard edge.

### 13.4 Exploit the Padé renormalization residual

The residual [equation \(8.14\)](#) is an explicit rational function. Several concrete questions are accessible computationally:

1. Does  $\mathfrak{E}_N(z)$  have a fixed sign on  $z > 1$  after a simple normalization?
2. Do its poles interlace with the affine images of the Gauss nodes?
3. Can the exact error formula turn [equation \(8.13\)](#) into two-sided recursive bounds for  $\mathcal{R}(z)$ ?
4. Does matching the Laurent expansion of  $\mathfrak{E}_N$  reproduce the moment recurrence in a triangular way, and can the first unmatched term be expressed solely through  $\beta_1, \dots, \beta_N$ ?

Positive answers would yield a structure-preserving numerical solver: compute the Jacobi truncation, enforce the dyadic residual, and obtain certified bounds without discretizing  $w$ .

### 13.5 Arithmetic of Hankel and Gaunt determinants

The dyadic-value literature studies denominators of  $F(2^{-n})$ . The native spectral data create a second arithmetic sequence:

$$\Delta_N, \quad D_N, \quad \beta_N, \quad \Pi_N. \quad (13.4)$$

Promising questions include:

1. determine exact 2-adic valuations of  $\Delta_N$  and  $\beta_N$ ;
2. isolate odd denominator factors coming from  $2^{2^r} - 1$  in the moment recurrence;
3. test whether cancellations in [equation \(7.7\)](#) admit finite valuation bounds uniform in  $N$ ;
4. express  $D_N$  directly through the rational Legendre coefficients  $c_0, \dots, c_{N-1}$  and classify the primes introduced by the Gaunt factorials.

Because  $\Pi_N \rightarrow \pi$ , any unexpectedly strong denominator control would have Diophantine consequences. No irrationality claim is made here; the immediate goal is an exact valuation theory parallel to the existing dyadic one.

### 13.6 The geometric $q$ -surface

For rational  $q$ , the whole moment/Jacobi/Gauss construction is exact. One can therefore sample the two-dimensional array  $\beta_n(q)$  without evaluating  $w_q$ . Specific targets are:

1. prove the sharp endpoint coefficient in [equation \(12.6\)](#);
2. find the  $q$ -analogue of [equation \(11.3\)](#);
3. determine whether  $n \mapsto \beta_n(q)$  is increasing for a full interval of  $q$  values;
4. study the limits  $q \downarrow 0$  (Wallis/Legendre) and  $q \uparrow 1$  after variance rescaling, where a Gaussian regime should emerge;
5. connect the cumulants [equation \(12.3\)](#) to the repository's  $q$ -Pochhammer and Gaussian-binomial filters.

This family provides a controlled deformation in which classical and Fabius phenomena can be compared rather than studied in isolation.

### 13.7 Numerical improvements

The monomial recurrence becomes costly beyond a few hundred degrees. Three improvements are recommended:

1. compute modified moments in the Chebyshev or Legendre basis and apply a modified Chebyshev algorithm for the Jacobi coefficients;
2. use interval or ball arithmetic to certify signs of  $1/4 - \beta_n$  and  $\beta_{n+1} - \beta_n$ ;
3. combine arbitrary precision with asymptotic preconditioning  $\beta_n = 1/4 - d_n$  to avoid subtractive loss in the tail.

The finite Gaunt formula [equation \(9.4\)](#) is particularly attractive because it replaces ill-conditioned monomial Hankel matrices by a near-orthogonal reference basis.

### 13.8 Lean formalization order

The repository already contains rational moments, positive Hankel matrices, low-degree native polynomials, and the Cauchy renormalization. It now also contains the finite scalar-naturality API and the all-degree rational native Jacobi layer catalogued in [section 4](#): exact  $\mathbb{Q}$ -valued determinants, polynomial coefficients, norms, zero diagonal, positive subdiagonals, determinant cross-ratio, orthogonality, centered recurrence, and real-cast bridges, with zero-based index  $n$  corresponding to  $\beta_{n+1}$ . The polynomial change-of-basis and Legendre specialization catalogued there now formalize [equations \(9.7\)](#) and [\(9.8\)](#), without formalizing the Gaunt/Wigner entry expansion. The remaining formalization sequence is:



1. **Finite algebra:** formalize [equations \(6.4\), \(7.2\), \(7.3\)](#) and [\(7.10\)](#). These need only finite linear algebra and the existing positivity and rational-native APIs. Separately, formalize [equations \(9.3\) to \(9.5\)](#) if an entrywise Gaunt/ $3j$ -symbol realization is desired.
2. **Quadrature and rational approximation:** define second-kind polynomials, prove [equation \(8.9\)](#), and derive the Gaussian weights and exactness theorem.
3. **Convexity:** add a reusable notion of log-concave measure, closure under convolution and weak convergence, then prove [theorem 3.1](#) and [corollaries 3.4](#) and [3.5](#).
4. **Classical asymptotic interfaces:** formalize or import abstract versions of Rakhmanov, strong Szegő, and Christoffel asymptotics. The Fabius-specific work then reduces to support, positivity, continuity, and the endpoint integrability estimate [equation \(5.3\)](#).
5. **Frontier rates:** only after the endpoint periodic normal form is proved should one attempt the conjectural  $n^{-2} \log^2 n$  correction.

The finite identities should not be blocked on the much larger formalization of OPRL asymptotics. In particular, the harmonic-mean theorem is already a useful exact target even if convergence to  $\pi$  remains an imported classical corollary initially.

*Remark 13.1* (Historical provenance). The pinned audit and novelty boxes above remain historical statements about their declared repository snapshot. This crosswalk updates current Lean status without retroactively altering those snapshot claims.

## 14 Conclusion

The repository’s existing theory describes the Fabius/Rvachev object through dyadic recursion, Thue–Morse cancellation, infinite sinc products, rational moments, special functions, and classical polynomial expansions. The present report adds a coherent intrinsic layer.

The probability law itself is log-concave. This converts the linear differential refinement equation into nonlinear Turán inequalities and strong convexity of the dyadic action. Its endpoint logarithmic square is integrable in the Szegő metric, forcing the native Jacobi matrix into the universal Nevai class. Christoffel functions then recover the bump from exact rational moment matrices. At the center, strong Szegő asymptotics collapse to an alternating product of rational recurrence coefficients whose limit is  $\pi$ . Gaussian quadrature and Padé approximation arise from the same finite Jacobi truncations, and the existing Legendre coefficients recover all of this data through finite Gaunt determinants.

The most striking closed chain is therefore

$$\boxed{\begin{array}{l} \text{dyadic Fabius arithmetic} \longrightarrow \text{rational moments} \longrightarrow \text{rational Jacobi data} \\ \longrightarrow \pi \longrightarrow \text{Christoffel reconstruction of up} \end{array}} \quad (14.1)$$

The chain is exact at every finite algebraic step and invokes classical analysis only for the limiting transitions. Its conjectural next layer—a  $2\pi^{-1}(\log_2 n)^2/n^2$  correction, dyadic spectral ripples, and a Lambert- $W$  endpoint kernel—is concrete enough for both high-precision experimentation and formal theorem development.

## A A detailed small-box estimate for the $q$ -family

This appendix supplies the estimate used in [equation \(12.5\)](#). Write

$$D_q = 1 - Y_q = 2(1 - q) \sum_{j=0}^{\infty} q^j Z_j, \quad (A.1)$$

where the  $Z_j = (1 - V_j)/2$  are independent uniform variables on  $[0, 1]$ . Choose

$$N = \left\lceil \frac{\log(4/t)}{|\log q|} \right\rceil, \quad (\text{A.2})$$

so that  $q^N \leq t/4$ . The tail after  $N$  is deterministically bounded by

$$2(1 - q) \sum_{j=N}^{\infty} q^j = 2q^N \leq \frac{t}{2}. \quad (\text{A.3})$$

On the event  $Z_0, \dots, Z_{N-1} \leq t/4$ , the prefix is at most

$$2(1 - q) \sum_{j=0}^{N-1} q^j \frac{t}{4} \leq \frac{t}{2}. \quad (\text{A.4})$$

Therefore

$$\Pr(D_q \leq t) \geq \left(\frac{t}{4}\right)^N, \quad (\text{A.5})$$

and hence

$$-\log \Pr(D_q \leq t) \leq N \log(4/t) = \mathcal{O}(\log^2(1/t)). \quad (\text{A.6})$$

Conditioning on the first uniform coordinate gives [equation \(12.4\)](#); replacing  $t$  by  $t/q$  changes only lower order constants. This proves the endpoint majorant needed for the Szegő integral. The estimate is intentionally crude: it proves the right scale class but not the sharp coefficient conjectured in [equation \(12.6\)](#).

## B Exact low-degree spectral table

The following entries are reproduced from the exact arithmetic output. Here  $c_n = m_{2n}$  and  $\Pi_M$  is defined by [equation \(7.4\)](#).

Table 3: Exact initial native spectral data.

| $n$ | $c_n$           | $H_n$                | $\beta_n$      | extra central datum         |
|-----|-----------------|----------------------|----------------|-----------------------------|
| 0   | 1               | 1                    | –              | –                           |
| 1   | 1/9             | 1/9                  | 1/9            | –                           |
| 2   | 19/675          | 32/2025              | 32/225         | $\Pi_1 = 64/25$             |
| 3   | 583/59535       | 19808/7441875        | 619/3675       | –                           |
| 4   | 132809/32531625 | 26727424/55791736875 | 835232/4640643 | $\Pi_2 = 53454848/19541211$ |

For the center Christoffel sequence, the exact relation

$$2M\lambda_{2M}(0) = \frac{M}{\sum_{m=0}^{M-1} \Pi_m^{-1}} \quad (\text{B.1})$$

gives

$$2\lambda_2(0) = 2, \quad 4\lambda_4(0) = \frac{128}{57}, \quad 6\lambda_6(0) = \frac{53454848}{22383145}. \quad (\text{B.2})$$

The CSV file continues exact moments, norms, and coefficients through degree 24; the later numerators and denominators are too large to improve a printed table.

## C Normalization checks against classical weights

Two model measures fix the constants in the strong asymptotic formulas.

### C.1 Lebesgue weight

For  $w(x) = 1$  on  $[-1, 1]$ , the monic polynomial is  $P_n/L_n$  and

$$H_n = \frac{2}{2n+1} \frac{1}{L_n^2} \sim \frac{\pi}{4^n}. \quad (\text{C.1})$$

Formula [equation \(5.9\)](#) gives

$$2\pi \exp\left(\frac{1}{\pi} \int_0^\pi \log \sin \theta \, d\theta\right) = 2\pi e^{-\log 2} = \pi, \quad (\text{C.2})$$

which matches [equation \(C.1\)](#).

### C.2 Uniform probability weight and the center constant

For  $w(x) = 1/2$ , the orthonormal polynomial is  $p_n(x) = \sqrt{2n+1} P_n(x)$ . The central asymptotic

$$P_{2M}(0) = (-1)^M \frac{\binom{2M}{M}}{4^M} \sim \frac{(-1)^M}{\sqrt{\pi M}} \quad (\text{C.3})$$

gives

$$p_{2M}(0)^2 \longrightarrow \frac{4}{\pi} = \frac{2}{\pi w(0)}, \quad (\text{C.4})$$

confirming [equation \(7.1\)](#). The corresponding recurrence coefficients are [equation \(12.9\)](#), and the general product [equation \(7.6\)](#) reduces to the Wallis product.

## D Computational method in pseudocode

The core exact/high-precision algorithm is short enough to state independently of the implementation.

**Algorithm D.1** (Moment-to-Jacobi recurrence). Given a target degree  $N$ :

1. Generate  $c_0, \dots, c_N$  by [equation \(2.11\)](#) and interleave  $m_{2j} = c_j$ ,  $m_{2j+1} = 0$ .
2. Initialize  $\pi_0 = 1$ ,  $\pi_1 = x$ ,  $H_0 = 1$ ,  $H_1 = m_2$ , and  $\beta_1 = H_1/H_0$ .
3. For  $n = 1, \dots, N-1$ :

$$\begin{aligned} \pi_{n+1} &= x\pi_n - \beta_n\pi_{n-1}, \\ H_{n+1} &= \sum_{j=0}^{n+1} [x^j] \pi_{n+1} m_{j+n+1}, \\ \beta_{n+1} &= H_{n+1}/H_n. \end{aligned}$$

4. Accumulate  $\Pi_M = 2 \prod_{j \leq M} \beta_{2j}/\beta_{2j-1}$  and then use [equation \(7.10\)](#) for the center Christoffel values.
5. Form the finite Jacobi matrix with off-diagonal entries  $\sqrt{\beta_n}$ ; its eigenvalues are the zeros of  $\pi_N$ .

The equality used in the norm step is [equation \(4.9\)](#). It avoids squaring a degree- $n$  polynomial, but it does not remove cancellation. Exact arithmetic is best for small  $n$ ; for large  $n$ , arbitrary precision must grow at least linearly with  $N$  in the current monomial implementation.

## E Corpus-audit notes

The nonduplication audit used the following hierarchy.

1. The current branch and commit were resolved first; the final snapshot is b863b90ee74ec0405af2d67ad04c61824c3aea00.
2. The recursive tree under `Analysis/FabiusFunction/docs` was inventoried, including canonical expositions, vendored papers, archives, generated TeX fragments, and semi-formalized frontier drafts.
3. The current coverage and audit documents were used to distinguish proved Lean declarations, report-level proofs, numerical claims, and unresolved conjectures.
4. The newest Legendre/Lagrange reports were inspected explicitly because they overlap thematically with polynomial reconstruction.
5. Targeted repository searches were made for *log-concavity*, *Christoffel*, *Szegő*, *Nevai*, *Rakhmanov*, *Gauss quadrature*, *J-fraction*, the alternating  $\beta_{2j}/\beta_{2j-1}$  product, and the Legendre–Gaunt determinant formulas.

No matching report-level result was found for the main claims developed here. The audit is necessarily corpus-relative: it does not establish that no analogous observation exists elsewhere in the worldwide literature.

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